

<<DOCUMENT MANAGEMENT FOR STUDYING WITH AI SYSTEM (AI STUDY HUB) >>

Software Requirement Specification

Group5:

Nguyễn Trần Duy - SE192263

Phạm Đức Hùng - SE182153

Quách Nguyễn Chí Hùng - SE192175

Nguyễn Anh Khoa - SE192067

Lê Đoàn Gia Anh - SE191043

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Table of Contents

1. INTRODUCTION	4
1.1. Purpose	4
1.2. Scope.....	4
1.3. Definitions, Acronyms, and Abbreviations	5
1.4. References.....	6
1.5. Overview.....	7
2. OVERALL DESCRIPTION.....	7
2.1. Product Perspective	7
2.1.1. System Architecture & Context Diagram.....	7
2.1.2. Use Case Diagram	8
2.1.3. User Interface Screen Flows	9
2.1.4. Conceptual Entity Relationship Diagram	11
2.1.5. Logical Entity Relationship Diagram (ERD)	11
2.1.6. Database Schema Diagram	14
2.1.1. System Context.....	15
2.1.2. Technology Stack	15
2.1.3. System Architecture	16
2.2. Product Functions	16
2.3. User Characteristics	17
2.4. Constraints	17
2.5. Assumptions and Dependencies	18
3. SPECIFIC REQUIREMENTS	18
3.1. Functional Requirements	18
3.1.1. Authentication & Account Management.....	18
3.1.2. User Profile & Account Management	20
3.1.3. Subject Management	20
3.1.4. Document Management.....	22
3.1.5. AI Chat (RAG — Retrieval-Augmented Generation).....	24
3.1.6. AI Quiz Generation & Management	25
3.1.7. AI Flashcard Generation & Management.....	26
3.1.8. Spaced Repetition Review (SM-2)	27
3.1.9. Gamification (XP / Level / Streak).....	28

3.1.10. Notifications	29
3.1.11. Payment & Tier Membership	30
3.1.12. Recommendations	33
3.1.13. Administration	33
3.2. Non-Functional Requirements.....	33
3.3. Interface Requirements.....	36
3.3.1. API Interface.....	36
3.3.2. WebSocket Interface.....	36
3.3.3. External System Interfaces	36
3.4. Data Requirements.....	36
3.4.1. Business Rules	36
4. SYSTEM HIGH-LEVEL DESIGN.....	40
4.1. System Architecture.....	40
4.2. Conceptual Entity Relationship Diagram	41
4.3. Logical Entity Relationship Diagram	41
4.3.1. User & Authentication Module	42
4.3.2. Content & AI Module.....	42
4.3.3. Gamification Module.....	42
4.3.4. Payment & Subscription Module	43
4.3.5. Enums (14).....	43
4.4. Database Design	43
4.4.1. Table Descriptions	43
5. USE CASES	46
5.1. Use Case Diagram	46
5.2. Use Case Descriptions	46
6. SYSTEM FUNCTIONS — SCREEN FLOW & AUTHORIZATION	48
6.1. Screen Flow	48
6.2. Screen Details	48
6.3. User Authorization.....	49
6.4. Non-Screen Functions	49
7. APPENDICES	51
Appendix A: API Endpoint Summary (23 Controllers + 1 SignalR Hub).....	51
Appendix B: Background Workers	52

Revision History

Version	Date	Author	Description
1.0	June 2026	Group 5	Initial SRS document
2.0	July 2026	Group 5	Revised to IEEE 830 standard; added NFRs, Spaced Repetition, Gamification, Recommendations, Real-time Notifications modules; corrected database design to match implementation (27 entities, 14 enums); fixed PayOS → VNPay; expanded Use Cases

1. INTRODUCTION

1.1. Purpose

This Software Requirements Specification (SRS) document comprehensively describes the functional requirements, non-functional requirements, and design scope for the **AI Study Hub** project (SWP391 — SE1923 — Group 5). It serves as the primary agreement between the development team and stakeholders, ensuring that all parties have a shared understanding of what the system will do, how it will perform, and how it will be validated.

Intended Audience: Project supervisors, development team members, frontend integration developers, QA testers, and system administrators.

1.2. Scope

AI Study Hub is an AI-integrated web platform designed to optimize the storage, search, and knowledge extraction of study materials. The system provides:

- **Centralized Management:** Secure cloud storage and systematic organization of documents by subject, significantly reducing document loss.
- **Optimized Search:** Hybrid semantic search (dense embeddings + BM25 sparse vectors) with Reciprocal Rank Fusion for intelligent document retrieval.
- **AI Integration:** RAG-powered chatbot for reading comprehension and direct Q&A, AI-generated quizzes and flashcards from document content.
- **Spaced Repetition:** SM-2 algorithm-based review scheduling to optimize long-term retention of flashcard content.
- **Gamification:** XP, levels, streaks, leaderboards, and badges to incentivize consistent study habits.
- **Convenient Sharing:** Document sharing between students with role-based access control.
- **Payment Integration:** VNPay gateway for premium tier subscriptions with expanded storage and AI token quotas.

Out of Scope: Mobile native applications, multi-language UI localization (backend only supports Vietnamese/English RAG responses), third-party LMS integration, real-time collaborative document editing.

1.3. Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification — this document
RAG	Retrieval-Augmented Generation — a technique that enhances LLM responses by retrieving relevant data from a knowledge base before generating an answer
LLM	Large Language Model — a deep learning model trained on large text corpora (e.g., GPT-4o-mini)
JWT	JSON Web Token — an open standard (RFC 7519) for stateless authentication via digitally signed tokens
OTP	One-Time Password — a single-use password for identity verification
OAuth	Open Authorization — a protocol for third-party delegated authorization (Google, GitHub)
CQRS	Command Query Responsibility Segregation — a pattern separating read and write operations
BM25	Best Matching 25 — a probabilistic ranking function for sparse text retrieval

RRF	Reciprocal Rank Fusion — a method to combine ranked result lists from multiple search strategies
SM-2	SuperMemo 2 — a spaced repetition algorithm that schedules review intervals based on recall quality
VNPay	A Vietnamese online payment gateway for processing financial transactions
EF Core	Entity Framework Core — Microsoft's Object-Relational Mapping (ORM) framework for .NET
SignalR	ASP.NET Core library for real-time bidirectional communication over WebSocket
Qdrant	An open-source vector similarity search engine used for semantic document retrieval
XP	Experience Points — a numerical measure of user activity within the gamification system
CRUD	Create, Read, Update, Delete — the four basic data manipulation operations
DTO	Data Transfer Object — an object used to transfer data between layers without exposing internal entities
MediatR	A .NET library implementing the Mediator pattern for in-process messaging (CQRS)

1.4. References

#	Document	Description
1	IEEE Std 830-1998	IEEE Recommended Practice for Software Requirements Specifications
2	SWP391 Course Syllabus	FPT University Software Development Project course requirements
3	ASP.NET Core 8 Documentation	Microsoft official documentation for the web framework
4	OpenAI API Reference	API documentation for embeddings and chat completions
5	Qdrant Documentation	Vector database REST API and configuration reference
6	VNPay Integration Guide	Payment gateway integration specification (HMAC-SHA512)
7	SM-2 Algorithm Paper	Piotr Wozniak, "SuperMemo 2 Algorithm" (1987)

1.5. Overview

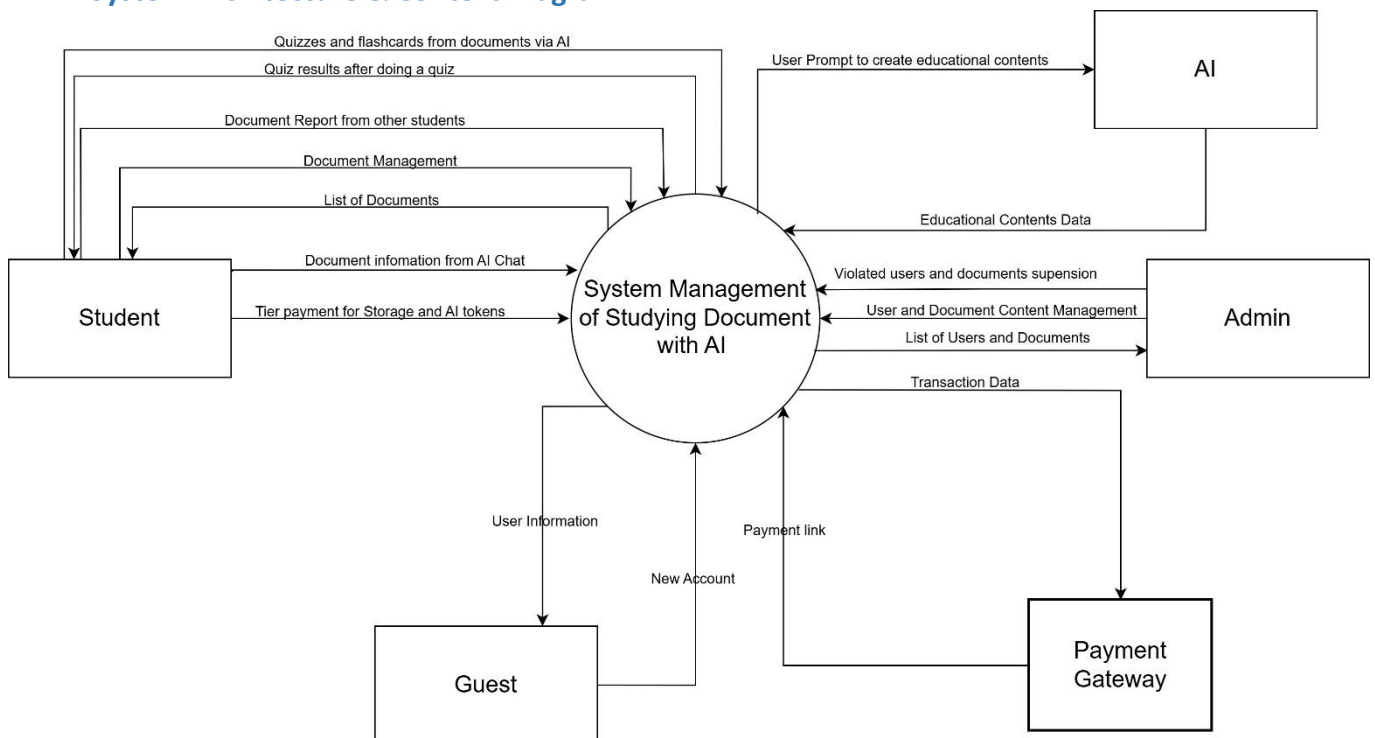
The remainder of this document is organized as follows:

- **Chapter 2** provides a high-level description of the product, its users, constraints, and assumptions.
- **Chapter 3** details every functional requirement (grouped by module), non-functional requirements, interface requirements, and data requirements.
- **Chapter 4** presents the system architecture, conceptual and logical ER diagrams, and complete database design.
- **Chapter 5** documents the use cases with descriptions.
- **Chapter 6** covers screen flows, screen details, user authorization matrices, and non-screen system functions.
- **Chapter 7** contains appendices including the API endpoint summary and background worker specifications.

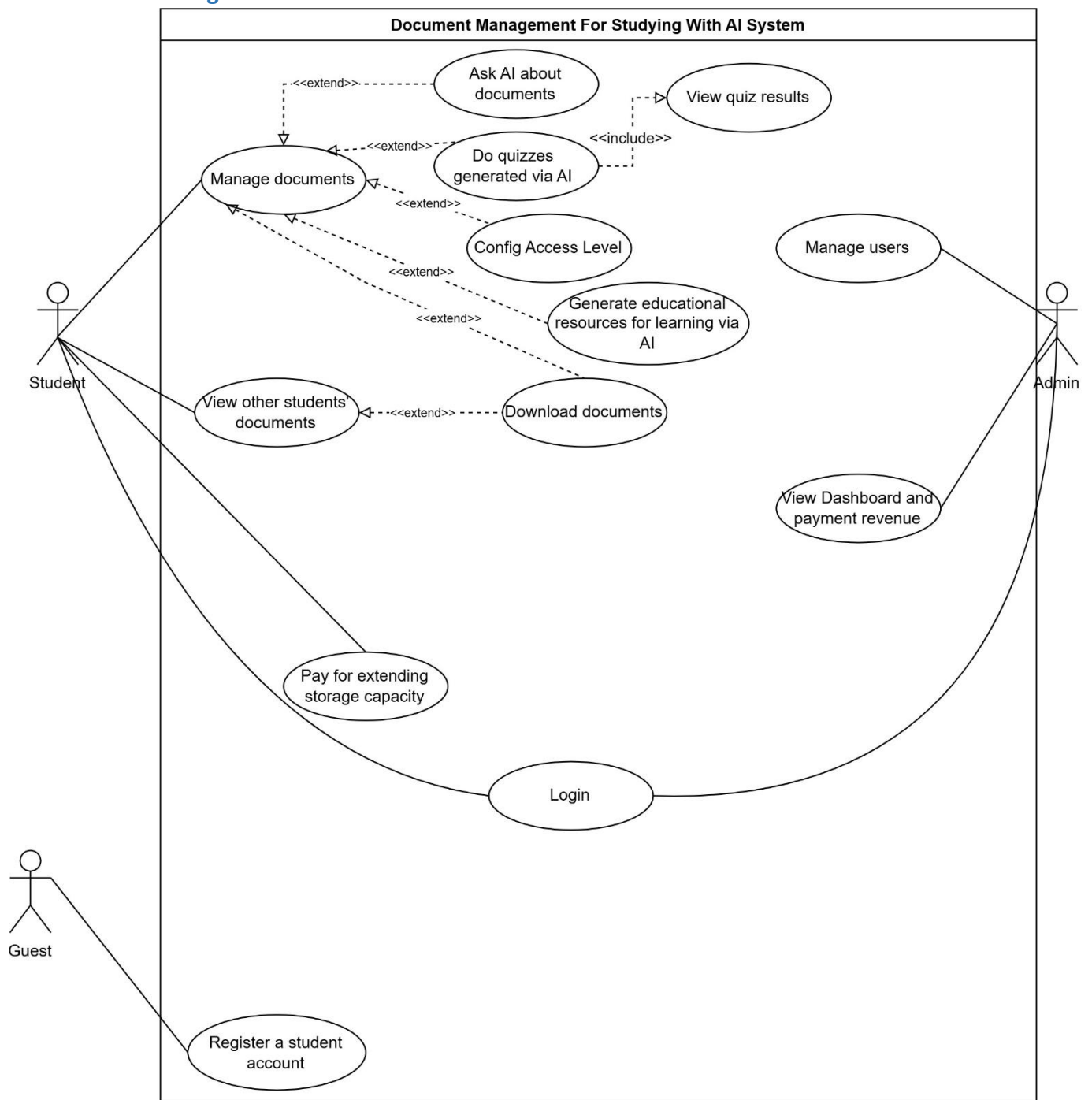
2. OVERALL DESCRIPTION

2.1. Product Perspective

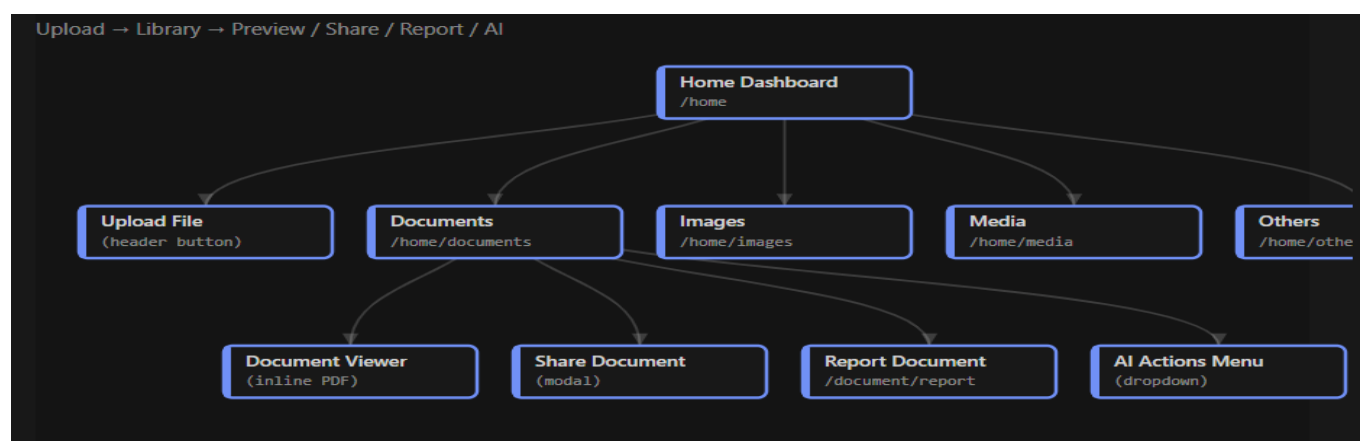
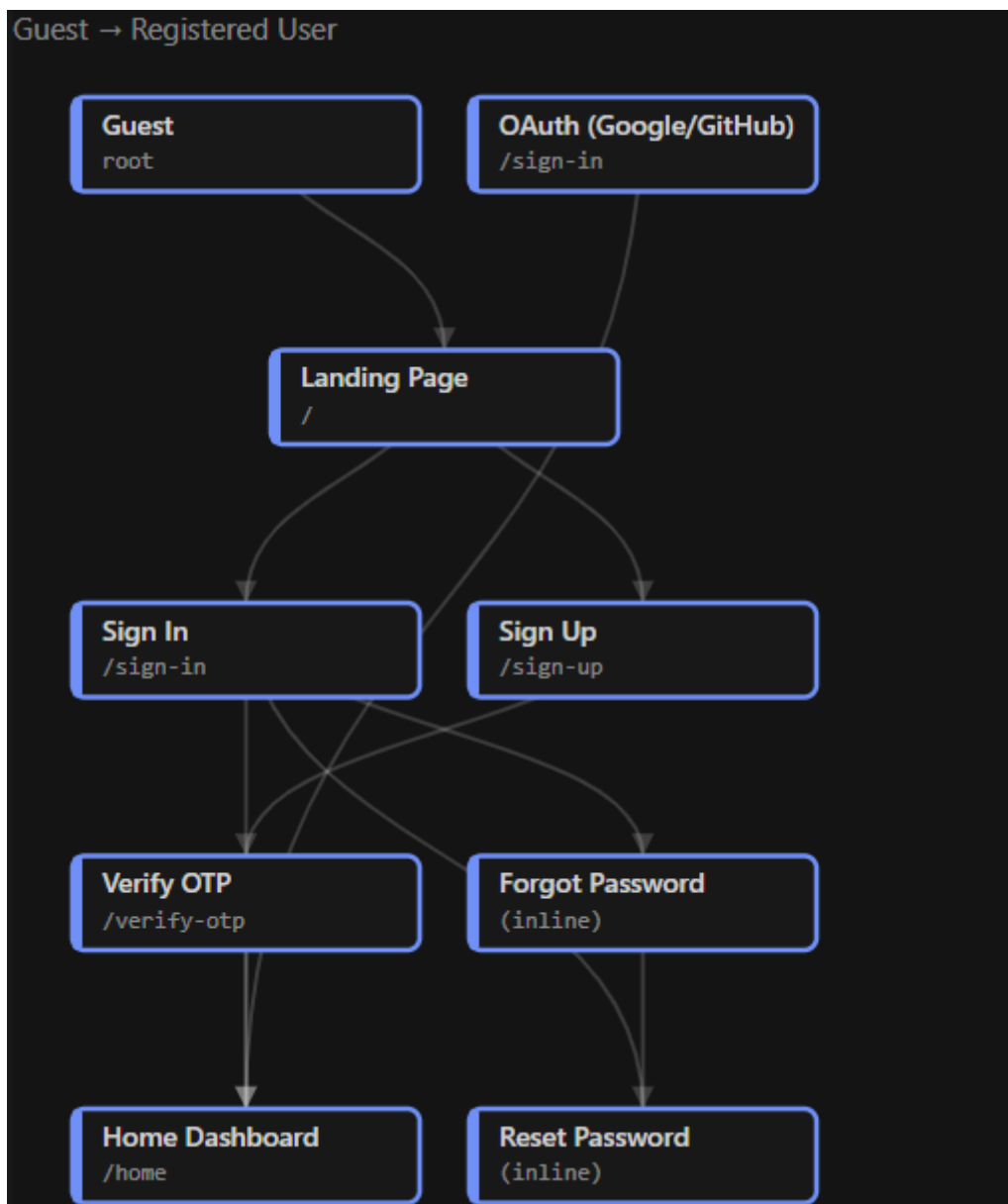
2.1.1. System Architecture & Context Diagram

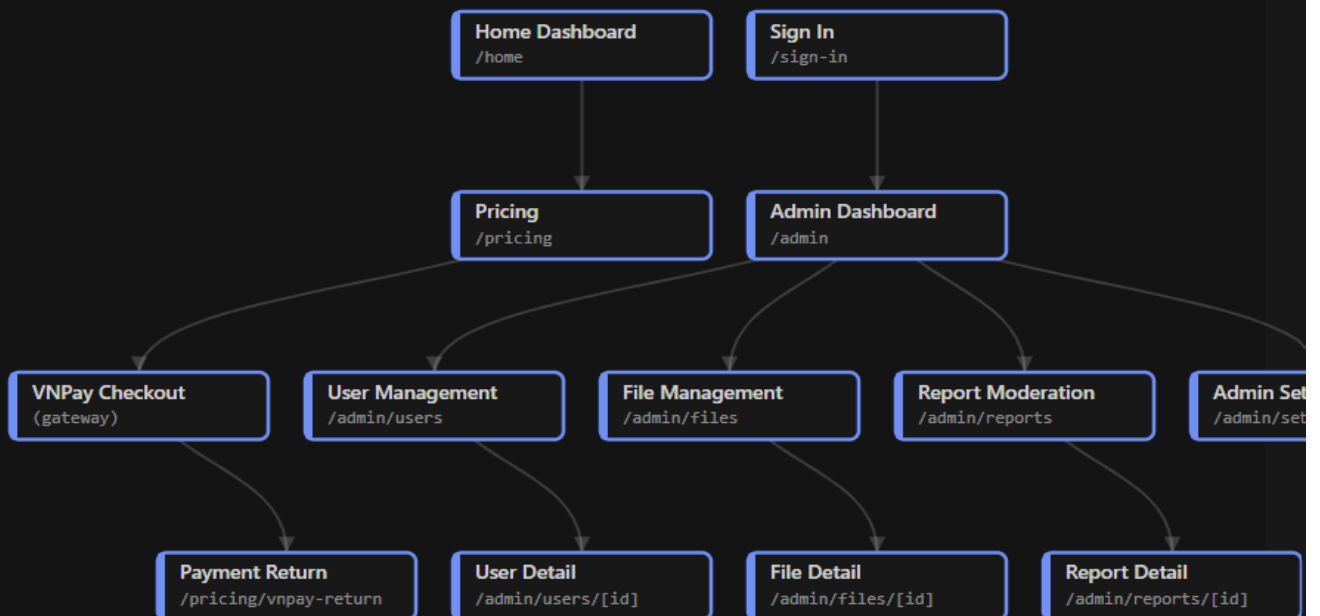
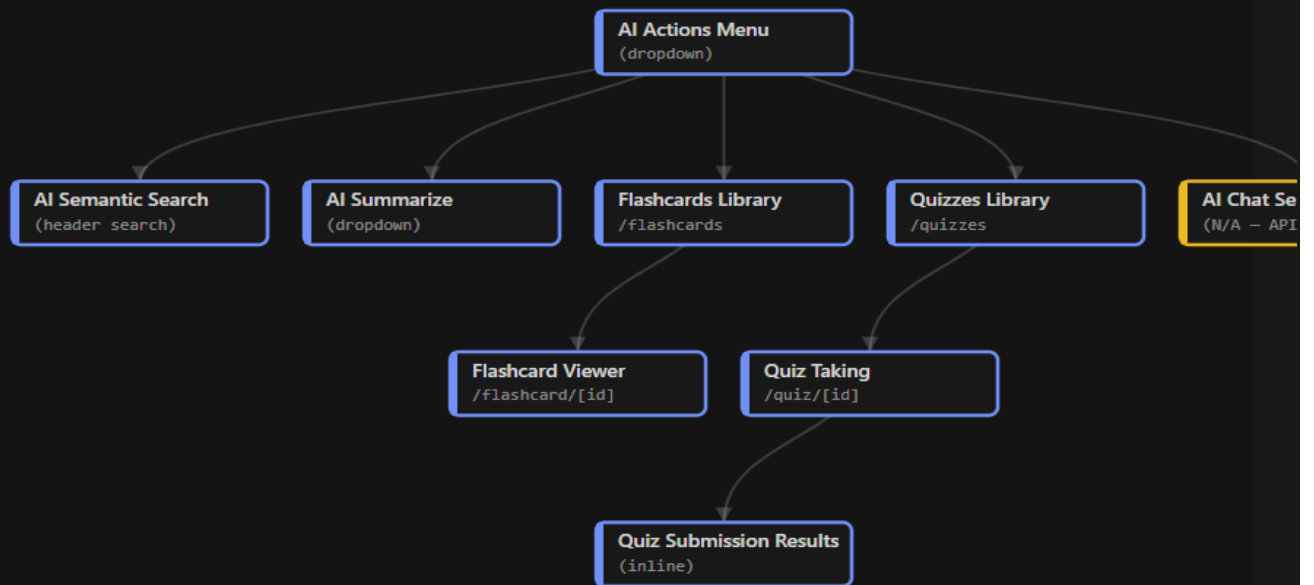


2.1.2. Use Case Diagram

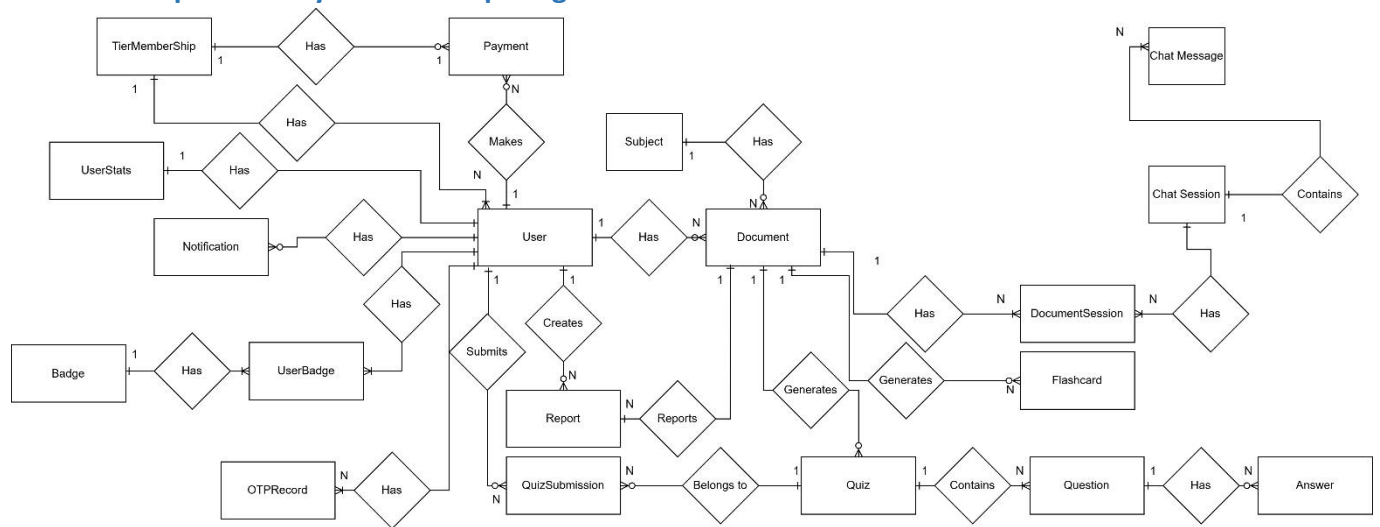


2.1.3. User Interface Screen Flows

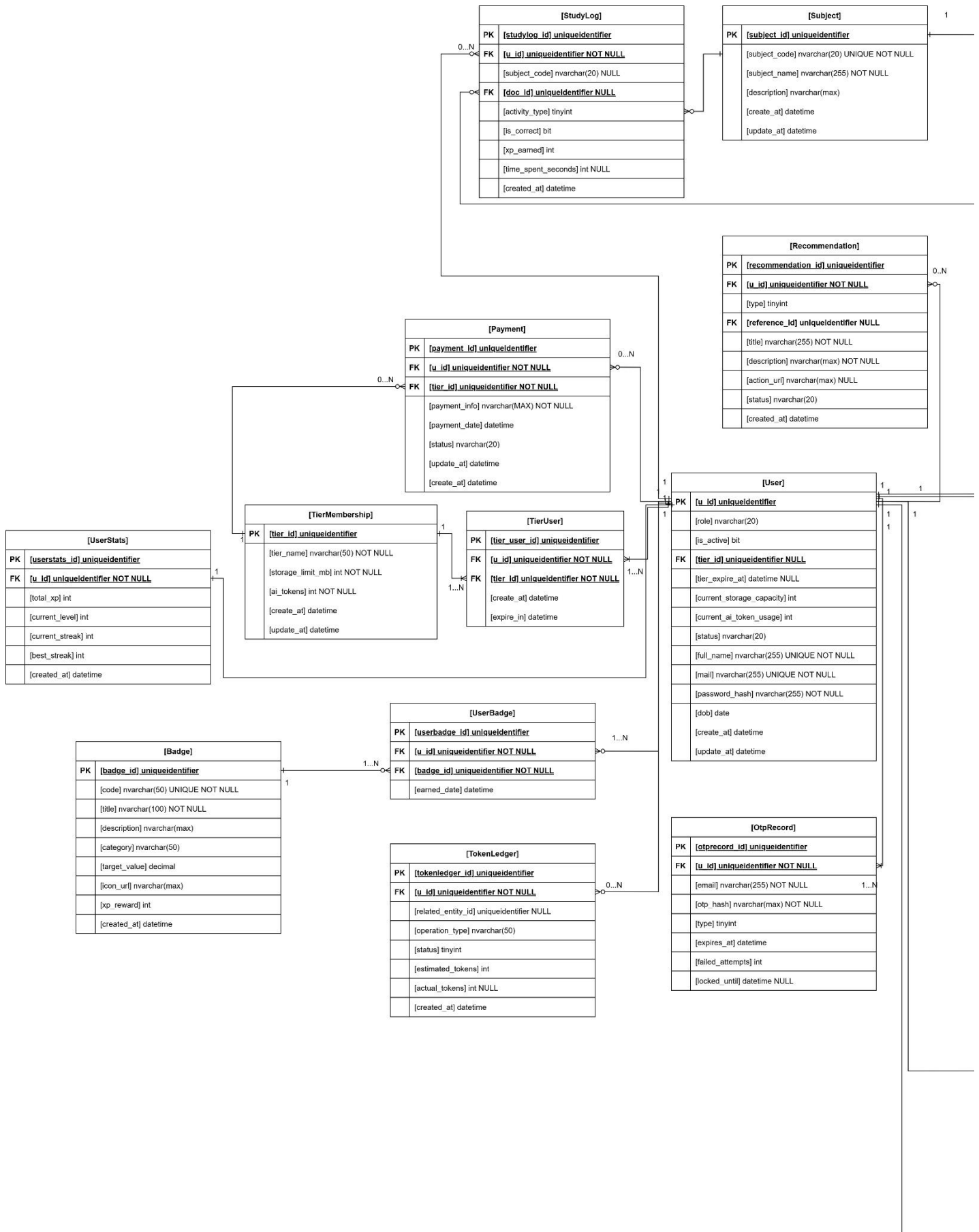


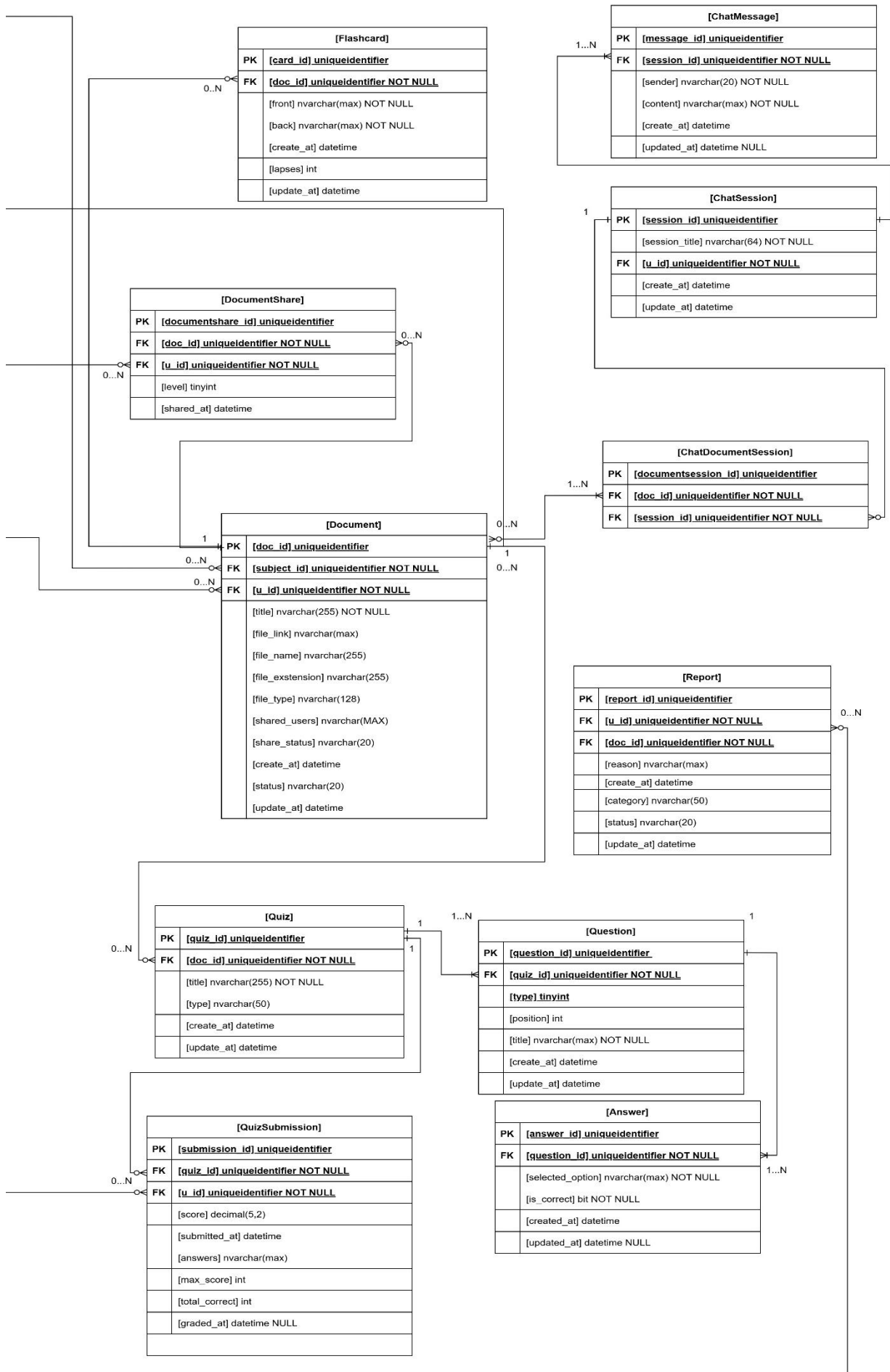


2.1.4. Conceptual Entity Relationship Diagram

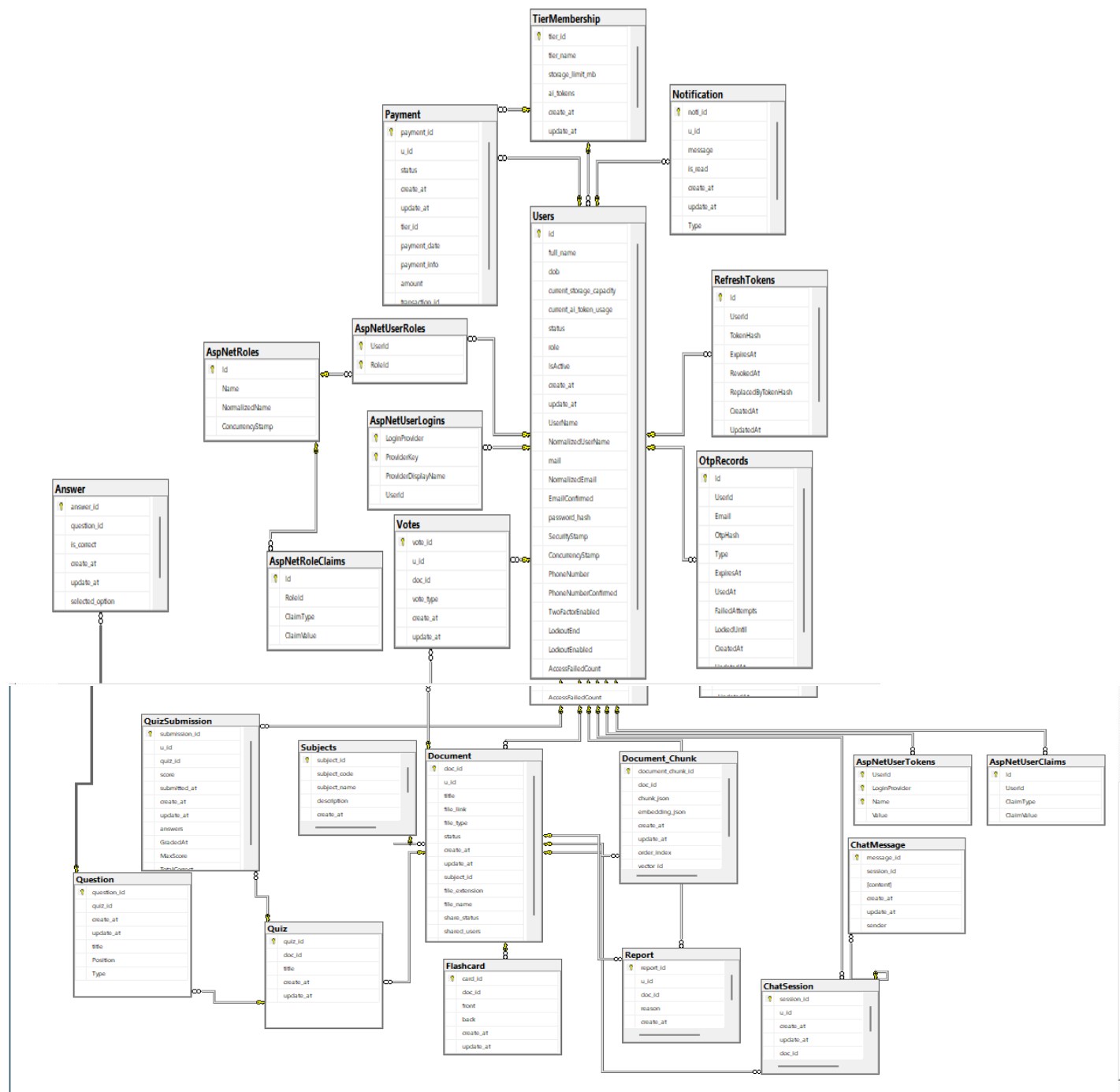


2.1.5. Logical Entity Relationship Diagram (ERD)





2.1.6. Database Schema Diagram



AI Study Hub is a standalone web application with a RESTful API backend. It integrates with external services but does not depend on any existing enterprise system.

2.1.1. System Context

External Entities:

Entity	Description
Guest	Unregistered user with limited access (viewing public previews, browsing categories, accessing pricing page)
User (Student)	Registered user who uploads/downloads documents, performs AI-powered searches, generates quizzes/flashcards, reviews with spaced repetition, and manages subscriptions
Admin	System administrator managing users, content moderation, tier memberships, and system-wide operations
OpenAI API	Third-party AI service providing text embeddings (text-embedding-3-small) and chat completions (gpt-4o-mini)
Qdrant	Local vector database storing document embeddings for semantic search
VNPay Gateway	Third-party financial platform processing user payments for tier upgrades via HMAC-SHA512 signed transactions
SMTP Server	Email delivery service for OTP codes and system notifications
Google OAuth	External identity provider for single sign-on via Google accounts
GitHub OAuth	External identity provider for single sign-on via GitHub accounts

2.1.2. Technology Stack

Category	Technology
Framework	ASP.NET Core 8 Web API
Language	C# 12
Database	SQL Server + Entity Framework Core 8 (Code First)
AI — Embedding	OpenAI SDK (text-embedding-3-small , 1536 dimensions)
AI — Vector Store	Qdrant (local, dense + sparse hybrid collections)
AI — LLM	OpenAI SDK (gpt-4o-mini)
AI — Orchestration	Semantic Kernel + Microsoft Kernel Memory
AI — Search	Hybrid search (dense + BM25 sparse, RRF fusion, positional decay reranking)
AI — Guardrails	FaithfulnessFilter, GroundingVerifier, ConfidenceScorer
Authentication	JWT Bearer + Refresh Tokens + OTP + Google OAuth + GitHub OAuth
ORM Mapping	AutoMapper
Validation	FluentValidation (with global action filter)

CQRS	MediatR (Auth, Users modules)
Logging	Serilog (console + rolling file)
Payment	VNPay (HMAC-SHA512 signature validation)
Real-time	SignalR WebSocket (/hubs/notifications)
Background Jobs	BackgroundService (DocumentProcessor, TierCleanup, UnverifiedCleanup, DailyStreakReset)
Patterns	Repository + Unit of Work, 3-Layer Architecture
Text Extraction	PdfPig (PDF), OpenXML (DOCX), raw reader (TXT/MD)

2.1.3. System Architecture

The system employs a strict 3-layer architecture:

Client

- AIStudyHub.API (Controllers, Middleware, JWT, Swagger, SignalR Hub, DI)
- AIStudyHub.Business (Services, AI pipeline, DTOs, Validators, Mappings, Workers)
- AIStudyHub.Data (EF Core DbContext, Repositories, UnitOfWork, Migrations, Seed)
- SQL Server + Qdrant

2.2. Product Functions

The system provides the following major functional areas:

#	Module	Description
1	Authentication	Register with email OTP verification, login with JWT + refresh tokens, password reset, Google/GitHub OAuth, logout
2	User Management	CRUD profiles, tier info, document sharing, admin user management
3	Document Management	Upload (PDF/DOCX/TXT/MD), text extraction, chunking, versioning, sharing, download, preview
4	RAG Pipeline	Hybrid search (dense + sparse BM25 via Qdrant RRF), reranking, Semantic Kernel orchestration, guardrails (faithfulness, grounding, confidence), document Q&A, summarization
5	AI Chat	Session-based chat with document context, RAG-augmented responses
6	AI Quiz Generation	AI-powered multiple-choice quiz creation from document chunks, auto-persisted with questions and answers

7	Quiz Submission	Submit answers, auto-grading, score tracking
8	AI Flashcard Generation	AI-powered flashcard (Q&A) creation from document chunks, auto-persisted
9	Spaced Repetition (SM-2)	Per-user flashcard review scheduling with ease factor, interval, and repetitions tracking
10	Gamification	XP, levels, streaks, leaderboard, activity logging via StudyLog, badges
11	Recommendations	Per-subject mastery analytics, AI-driven study suggestions
13	Notifications	DB-backed notification list + real-time push via SignalR WebSocket
14	Payments	VNPay checkout, webhook processing, tier upgrade on success
15	Tier Memberships	Subscription tiers with storage and AI token quotas
16	Administration	Dashboard, document reindexing, user moderation

2.3. User Characteristics

Role	Description	Technical Proficiency
Guest	Unauthenticated visitor exploring the platform; can view public pages, pricing, and initiate registration	Low — basic web browsing skills
Registered User (Student)	University student who uploads study materials, uses AI tools, reviews flashcards, takes quizzes, and manages subscriptions	Medium — comfortable with web applications, document management, and online payment
Admin	System administrator monitoring platform health, moderating content, managing users, and configuring system settings	High — familiar with admin dashboards, API testing, and system configuration

2.4. Constraints

#	Constraint
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C-01	The system must be developed using ASP.NET Core 8 and C# 12 as mandated by the SWP391 course
C-02	SQL Server is the required relational database engine
C-03	The AI pipeline requires access to the OpenAI API (API key required)
C-04	Qdrant vector database must be running locally on port 6333
C-05	Payment processing is limited to VNPay gateway
C-06	File uploads are limited to .pdf, .docx, .txt, and .md formats with a maximum size of 50 MB per file
C-07	The system must support Vietnamese as the default language for AI responses

2.5. Assumptions and Dependencies

#	Assumption/Dependency
A-01	The OpenAI API will remain available and backward-compatible during development and deployment
A-02	Users have modern web browsers (Chrome, Firefox, Edge, Safari) with WebSocket support for SignalR
A-03	SQL Server and Qdrant are provisioned and accessible in the deployment environment
A-04	SMTP server credentials are configured for OTP email delivery
A-05	VNPay sandbox/production credentials are available for payment testing
A-06	Google and GitHub OAuth client IDs/secrets are registered and configured
A-07	The development team has access to .NET 8 SDK and standard development tools

3. SPECIFIC REQUIREMENTS

3.1. Functional Requirements

3.1.1. Authentication & Account Management

FR ID	Function Name	Trigger / API Endpoint	Description	Abnormal Cases
FR-1.1	Register New Account	POST /api/Auth/register	Creates a new Identity user account with FullName, Email, Password, and optional DateOfBirth. Generates and emails a 6-digit OTP. Account	Invalid input → 400; duplicate email → validation error

			created in unverified state. Auto-assigns Student role and Free tier. Email/password validated by FluentValidation.	
FR-1.2	Verify Registration OTP	POST <code>/api/Auth/verify-registration-otp</code>	Compares submitted OTP against stored record. On match, sets <code>EmailConfirmed = true</code> . OTP must be valid and not expired (3-minute window).	Expired/mismatched OTP → validation error
FR-1.3	Resend Registration OTP	POST <code>/api/Auth/resend-registration-otp</code>	Invalidates previous OTP and generates a new one, emailed to the user. Only for unverified accounts. Returns 200 OK regardless of email existence (anti-enumeration).	—
FR-1.4	Login	POST <code>/api/Auth/login</code> (rate limited: 5 req/15 min per IP)	Authenticates via ASP.NET Core Identity, issues JWT access token + refresh token. <code>EmailConfirmed</code> must be true. Returns <code>AuthResponseDto</code> with user profile, tokens, expiry.	Invalid credentials → 400/401; unverified email → 401; rate limit → 429
FR-1.5	Refresh Access Token	POST <code>/api/Auth/refresh-token</code>	Validates the refresh token, issues new access token with refresh token rotation. Old refresh token is invalidated.	Invalid/expired/revoked token → 401
FR-1.6	Forgot Password	POST <code>/api/Auth/forgot-password</code>	Sends a password-reset OTP to user's email. Response is identical regardless of email existence (anti-enumeration).	—
FR-1.7	Reset Password	POST <code>/api/Auth/reset-password</code>	Validates OTP, updates password, revokes all existing refresh tokens for security. New password must satisfy complexity rules.	Invalid OTP or weak password → validation error
FR-1.8	Change Password	POST <code>/api/Auth/change-password</code> (Authenticated)	Validates current password before applying new one. New password must differ from current.	Wrong current password → 400/401
FR-1.9	Logout	POST <code>/api/Auth/logout</code>	Revokes the supplied refresh token so it can no longer be used to obtain new access tokens.	—
FR-1.10	External Login (Google / GitHub)	GET <code>/api/Auth/external-login/{provider}</code>	Redirects to external OAuth provider. Callback creates local user on first use with auto-confirmed email, Student role,	Unsupported provider → 400; failed auth → 401

		→ redirect → callback	Free tier. Returns AuthResponseDto .	
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3.1.2. User Profile & Account Management

FR ID	Function Name	Trigger / API Endpoint	Description	Abnormal Cases
FR-2.1	Get Current User Tier	GET /api/User/me/tier	Returns tier info (name, storage limit, AI-token quota, expiry date) for the authenticated user.	Unauthenticated → 401
FR-2.2	Update My Profile	PUT /api/User/me	Updates profile fields (full name, avatar URL). Email and password are not modifiable through this endpoint. Returns 204 NoContent.	—
FR-2.3	Search Shareable Users	GET /api/User/shareable?keyword=...	Returns users (id, name, email) that can receive shared documents, filtered by keyword. Current user is always excluded from results.	—
FR-2.4	Admin: List All Users	GET /api/User (Admin only)	Returns the full list of registered users with profile information.	Not admin → 403
FR-2.5	Admin: Get User by ID	GET /api/User/{id} (Admin only)	Returns a single user's profile by their identifier.	Not found → 404; not admin → 403
FR-2.6	Admin: Update User Tier	PUT /api/User/{id}/tier (Admin only)	Assigns a new tier membership to the specified user, optionally setting an expiration date. Returns 204 NoContent.	Not admin → 403

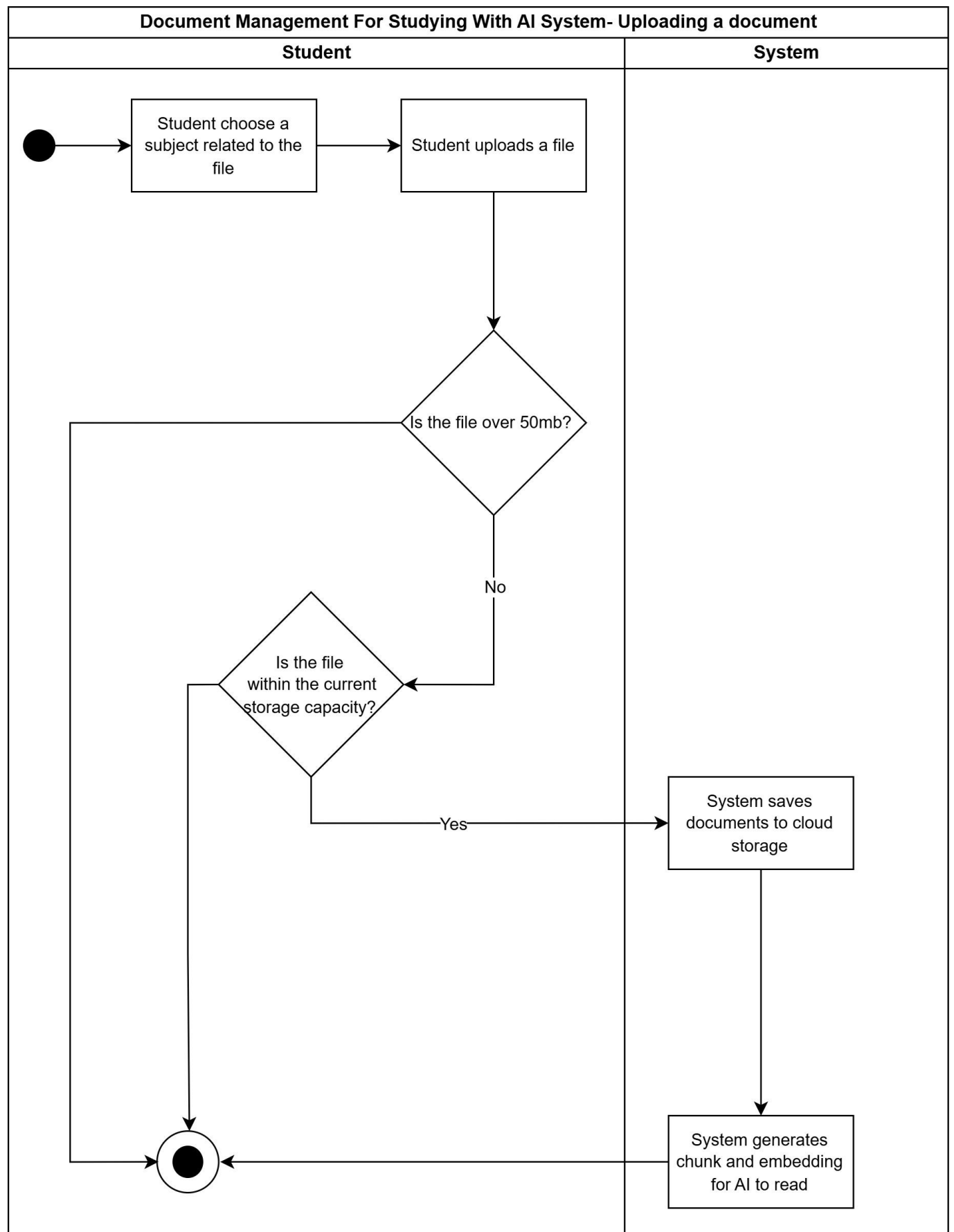
3.1.3. Subject Management

FR ID	Function Name	Trigger / API Endpoint	Description	Abnormal Cases
FR-3.1	List Subjects	GET /api/Subject	Returns a paginated list of subjects. Supports PaginationParams .	—
FR-3.2	Get Subject by ID	GET /api/Subject/{id}	Returns a single subject by its identifier.	Not found → 404

FR-3.3	Create Subject	POST <code>/api/Subject</code> (Student or Admin)	Creates a new subject with SubjectCode (unique), SubjectName, Description. Returns 201 Created.	Duplicate SubjectCode → validation error
FR-3.4	Update Subject	PUT <code>/api/Subject/{id}</code> (Admin only)	Updates the subject's name or description.	Not admin → 403
FR-3.5	Delete Subject	DELETE <code>/api/Subject/{id}</code> (Admin only)	Hard-deletes the subject. Documents must be re-categorized beforehand. Returns 204 NoContent.	Not admin → 403

3.1.4. Document Management

Main-Flow Diagram: Document Uploading & Processing



FR ID	Function Name	Trigger / API Endpoint	Description	Abnormal Cases
FR-4.1	Upload Document	POST <code>/api/Document/upload</code> (multipart/form-data)	Accepts .pdf, .docx, .txt, .md files. Validates user, subject, file size (max 50 MB), and storage quota. Creates Document with Status = Processing , ShareStatus = private . Enqueues for background AI processing. Returns 202 Accepted.	Missing file → 400; unknown subject → 400; quota exceeded → 403; unsupported extension → 400
FR-4.2	Get Upload Status	GET <code>/api/Document/{id}/status</code>	Returns current processing status (Processing, Published, Failed). Owner only.	Not owner → 403; not found → 404
FR-4.3	List Documents	GET <code>/api/Document?subjectId=...</code>	Paginated list of user's own documents + public/shared documents. Supports subject filtering and PaginationParams .	—
FR-4.4	Get Document by ID	GET <code>/api/Document/{id}</code>	Returns a single document's metadata.	Not found → 404
FR-4.5	Update Document Metadata	PUT <code>/api/Document/{id}</code>	Updates title, file metadata, ShareStatus. Does not modify the underlying file or its vector embeddings.	Not found → 404
FR-4.6	Share Document	POST <code>/api/Document/{id}/share</code>	Sets shared user IDs list. ShareStatus auto-derived: non-empty list → "shared", empty → "private". Owner only. Self-sharing is filtered out.	Not owner → 403
FR-4.7	Delete Document Metadata	DELETE <code>/api/Document/{id}</code>	Removes document metadata from the database only. File, chunks, and vectors are not affected. Returns 204 NoContent.	—
FR-4.8	Download Document File	GET <code>/api/Document/{id}/download</code>	Streams the physical file with Content-Disposition: attachment header for browser download.	Not found → 404
FR-4.9	Preview Document File	GET <code>/api/Document/{id}/preview</code>	Streams the physical file with Content-Disposition: inline header for browser rendering.	Not found → 404
FR-4.10	Get Document Chunks	GET <code>/api/Document/{id}/chunks</code>	Returns up to 1,000 text chunks from vector store. Used for debugging and showing AI source snippets.	Not found → 404

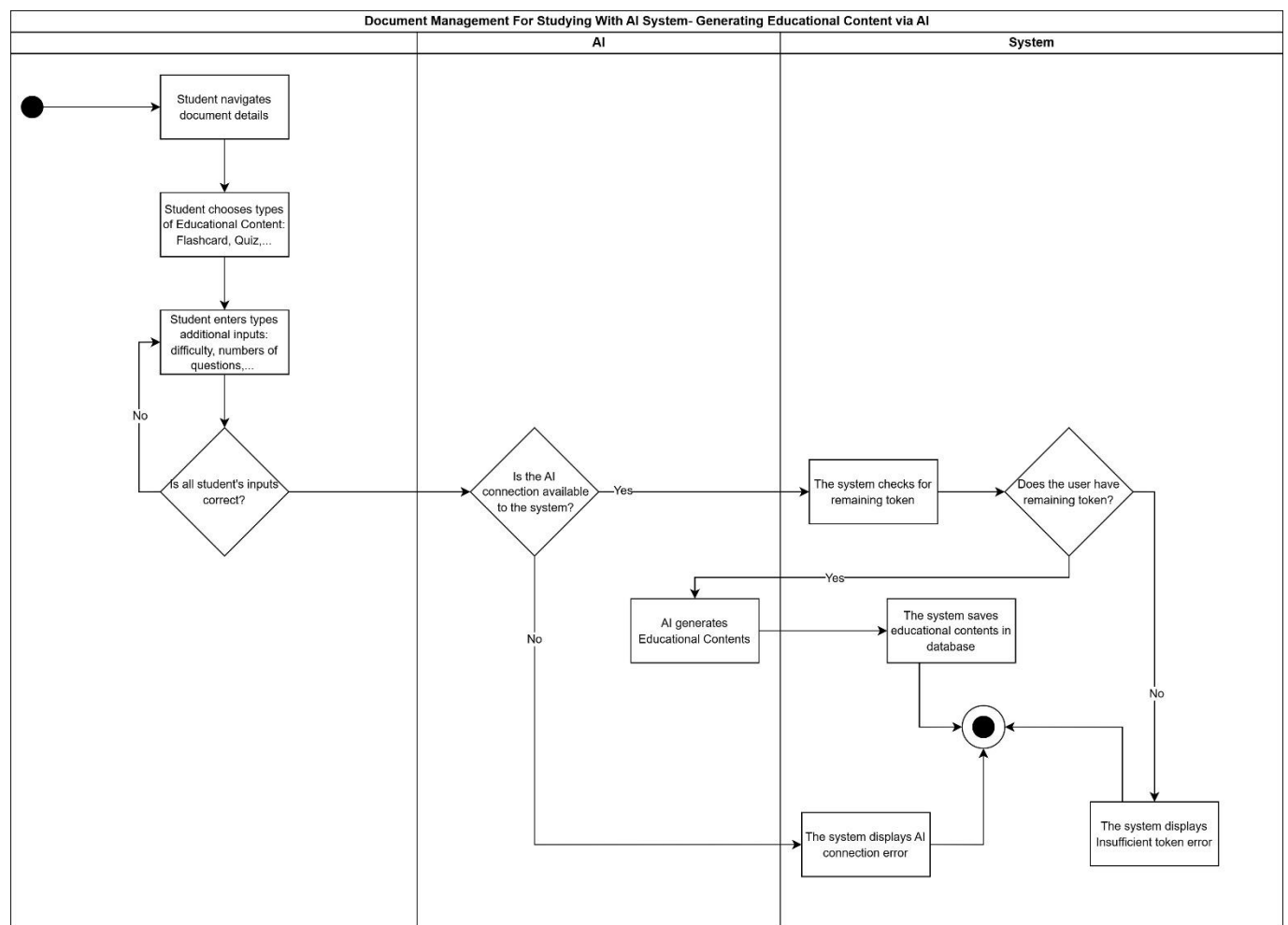
FR-4.11	Re-process Document	POST /api/Document/{id}/reprocess	Deletes existing vectors, resets status to Processing, re-enqueues for background processing. Owner only. Returns 202 Accepted.	Not owner → 403; file missing → 400
FR-4.12	Delete Document (Full)	DELETE /api/Document/{id}/full	Full delete: vectors, database record, physical file, storage quota decrement. Owner only. Returns 204 NoContent.	Not owner → 403; not found → 404

3.1.5. AI Chat (RAG — Retrieval-Augmented Generation)

FR ID	Function Name	Trigger / API Endpoint	Description	Abnormal Cases
FR-5.1	Ask a Question (RAG)	POST /api/AI/rag/ask	Embeds question, performs hybrid search (dense + sparse) against user's vectors, reranks top 5 chunks, calls LLM with system prompt, applies guardrails (faithfulness, grounding, confidence). Returns { answer , citations , confidence }.	Empty question → 400; unauthenticated → 401
FR-5.2	Summarize Document	POST /api/AI/rag/summarize	Retrieves document chunks and asks LLM to produce a concise summary. Returns { summary }.	Unauthenticated → 401; server error → 500
FR-5.3	List Chat Sessions	GET /api/Chat/sessions	Returns all chat sessions belonging to the authenticated user in chronological order.	—
FR-5.4	Create Chat Session	POST /api/Chat/sessions	Creates a new empty chat session for the authenticated user. Returns ChatSessionResponseDto .	—
FR-5.5	Get Chat Messages	GET /api/Chat/sessions/{sessionId}/messages	Returns all messages in the given session in chronological order.	—
FR-5.6	Send Chat Message	POST /api/Chat/messages	Persists user message, sends to RAG pipeline, persists AI response and citations. Returns ChatMessageResponseDto .	—

3.1.6. AI Quiz Generation & Management

Main-Flow Diagram: Generating Educational Content

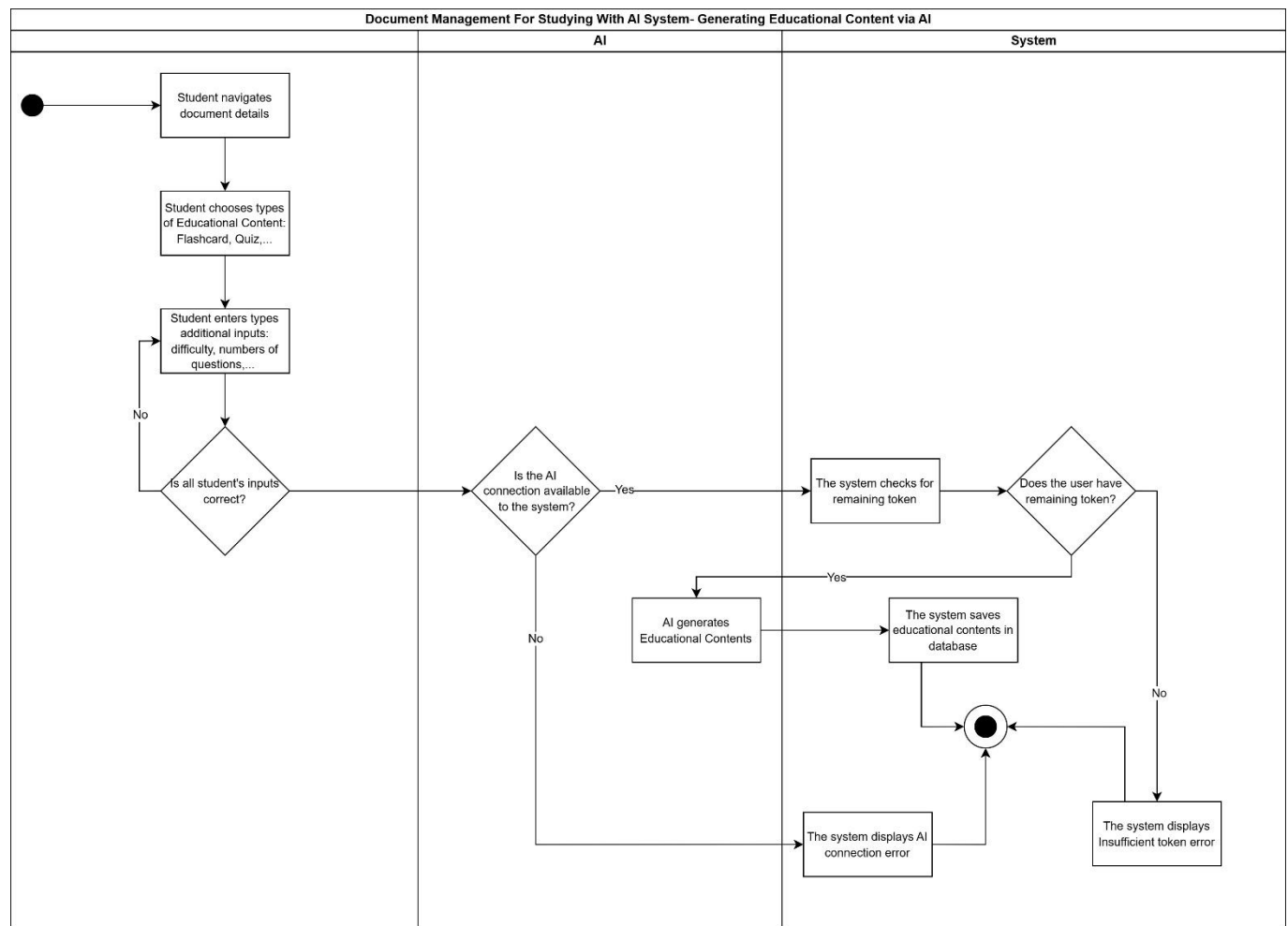


FR ID	Function Name	Trigger / API Endpoint	Description	Abnormal Cases
FR-6.1	Generate Quiz (AI)	POST /api/AI/quizzes/generate	Reads document chunks, asks LLM to generate 1–20 multiple-choice questions with 4 options each (exactly 1 correct). Persists Quiz → Questions → Answers hierarchy. Batch size 3, max 60 attempts, 3-stage JSON parser.	Not found → 404; out-of-range count → 400; not owner → 403
FR-6.2	List Quizzes	GET /api/Quiz	Returns paginated list of quizzes. Supports PaginationParams .	—
FR-6.3	Get Quiz by ID	GET /api/Quiz/{id}	Returns a single quiz including its questions and answers.	Not found → 404
FR-6.4	Create Quiz Manually	POST /api/Quiz	Creates quiz from CreateQuizRequestDto . Legacy endpoint — AI generation preferred. Returns 201 Created.	—

FR-6.5	Update Quiz	PUT /api/Quiz/{id}	Updates quiz title. Returns 200 OK.	Not found → 404
FR-6.6	Delete Quiz	DELETE /api/Quiz/{id}	Deletes quiz with all child questions and answers. Returns 204 NoContent.	Not found → 404
FR-6.7	Question & Answer CRUD	Quiz sub-resources	Full CRUD for Questions and Answers via quiz sub-resource endpoints. Legacy individual endpoints remain for backward compatibility.	Not found → 404
FR-6.8	Quiz Submission	GET /api/QuizSubmission , GET /api/QuizSubmission/{id}	Read-only access to quiz submissions. List is admin-only. Submit + scoring handled as business action on quiz domain.	Not admin (list) → 403

3.1.7. AI Flashcard Generation & Management

Main-Flow Diagram: Generating Educational Content



FR ID	Function Name	Trigger / API Endpoint	Description	Abnormal Cases
FR-7.1	Generate Flashcards (AI)	POST <code>/api/AI/flashcards/generate</code>	Reads chunks via Kernel Memory (max 30k chars), generates Q&A flashcard pairs via LLM (batch size 5, max 80 attempts). Deduplicates by normalized front text, enforces front-as-question/back-as-answer. Owner only.	Not owner → 403
FR-7.2	Get Flashcards by Document	GET <code>/api/Flashcard/document/{docId}</code>	Returns all flashcards previously generated for the given document.	—
FR-7.3	List Flashcards	GET <code>/api/Flashcard</code>	Returns paginated list of all flashcards. Supports <code>PaginationParams</code> .	—
FR-7.4	Get Flashcard by ID	GET <code>/api/Flashcard/{id}</code>	Returns a single flashcard.	Not found → 404
FR-7.5	Create Flashcard Manually	POST <code>/api/Flashcard</code>	Creates flashcard from <code>CreateFlashcardRequestDto</code> . Legacy — AI generation preferred. Returns 201.	—
FR-7.6	Update Flashcard	PUT <code>/api/Flashcard/{id}</code>	Updates flashcard content (front/back).	Not found → 404
FR-7.7	Delete Flashcard	DELETE <code>/api/Flashcard/{id}</code>	Deletes a single flashcard. Returns 204 NoContent.	Not found → 404

3.1.8. Spaced Repetition Review (SM-2)

FR ID	Function Name	Trigger / API Endpoint	Description	Abnormal Cases
FR-8.1	Get Due Flashcards	GET <code>/api/FlashcardReview/due?limit=20</code>	Returns flashcards whose <code>NextReviewDate ≤ UtcNow</code> for the authenticated user, ordered by date, limited by the <code>limit</code> parameter.	—
FR-8.2	Submit Review	POST <code>/api/FlashcardReview/review</code>	Processes a flashcard review with quality score (0–5). Applies SM-2 algorithm: $EF' = \max(1.3, EF$	—

			$+ (0.1 - (5-q) \times (0.08 + (5-q) \times 0.02)))$. Quality ≥ 3 : interval progresses ($1 \rightarrow 6 \rightarrow \text{prev} \times \text{EF}$), repetitions++. Quality < 3 : interval=1, repetitions=0. $\text{NextReviewDate} = \text{UtcNow} + \text{interval days}$. Upserts FlashcardReview record. Awards XP via GamificationService.	
FR-8.3	Get Due Count	GET <code>/api/FlashcardReview/due/count</code>	Returns the number of flashcards due for review today for the authenticated user.	—
FR-8.4	Get User Review Stats	GET <code>/api/FlashcardReview/stats</code>	Returns review statistics: total reviewed, mastery distribution, streak info for the authenticated user.	—

3.1.9. Gamification (XP / Level / Streak)

FR ID	Function Name	Trigger / API Endpoint	Description	Abnormal Cases
FR-9.1	Get My Stats	GET <code>/api/Gamification/stats</code>	Returns UserStats : TotalXp, CurrentLevel, CurrentStreak, BestStreak, LastActivityDate for the authenticated user.	—
FR-9.2	Get Leaderboard	GET <code>/api/Gamification/leaderboard</code>	Returns users ranked by TotalXp (descending). Shows level and streak info alongside.	—
FR-9.3	Award XP (Internal)	POST <code>/api/Gamification/award-xp</code> (server-to-server)	Awards XP: $\text{totalXp} += \text{xpDelta}$; $\text{level} = 1 + \text{floor}(\text{totalXp}/100)$. Streak logic: consecutive day $\rightarrow \text{currentStreak}++$, skip $\rightarrow$ reset to 1. $\text{bestStreak} = \text{max}(\text{bestStreak}, \text{currentStreak})$. Inserts StudyLog (append-only). Level-up $\rightarrow$ pushes SignalR notification.	—
FR-9.4	View Activity History	GET <code>/api/Gamification/activity</code>	Returns the user's StudyLog entries (ActivityType, DocumentId, SubjectCode,	—

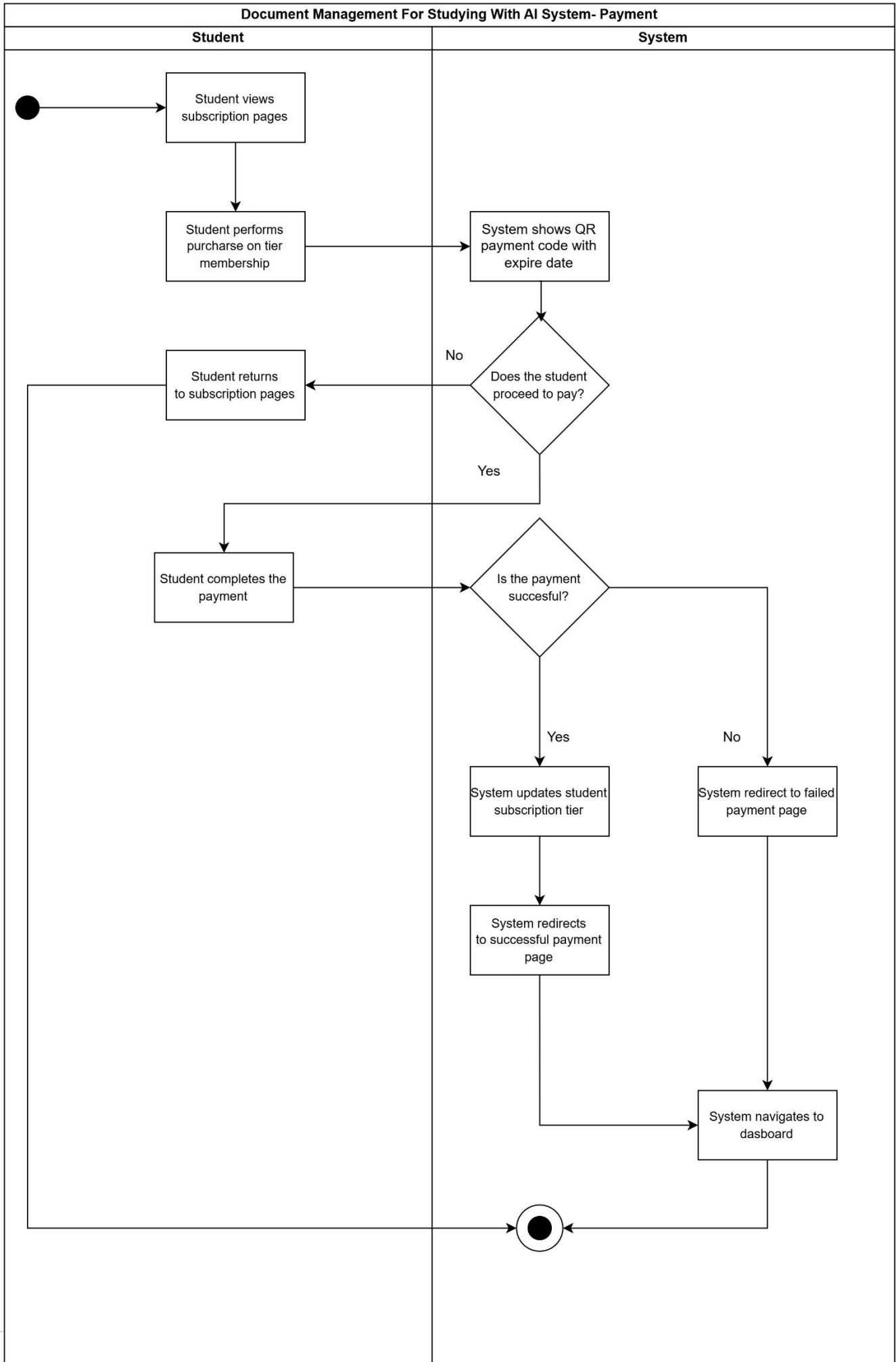
			IsCorrect, TimeSpentSeconds, XpEarned, timestamps).	
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3.1.10. Notifications

FR ID	Function Name	Trigger / API Endpoint	Description	Abnormal Cases
FR-10.1	List My Notifications	GET <code>/api/Notification/my</code>	Returns all notifications addressed to the authenticated user.	—
FR-10.2	Get Notification by ID	GET <code>/api/Notification/{id}</code>	Returns a single notification.	Not found → 404
FR-10.3	Mark as Read	POST <code>/api/Notification/{id}/read</code>	Sets the notification's IsRead flag to true. Returns 204 NoContent.	—
FR-10.4	Mark All as Read	POST <code>/api/Notification/mark-all-read</code>	Marks every notification belonging to the authenticated user as read. Returns 204 NoContent.	—
FR-10.5	Admin: List All	GET <code>/api/Notification</code> (Admin only)	Returns every notification in the system.	Not admin → 403
FR-10.6	Real-time Push (SignalR)	<code>ws(s)://host/hubs/notifications?access_token=<JWT></code>	Clients connect via WebSocket, join group by userId. Receive push for: Document ready, Quiz/Flashcards ready, StreakAtRisk, TierUpgraded, Payment events. Events: <code>JoinGroup(userId)</code> , <code>LeaveGroup(userId)</code> , <code>ReceiveNotification(payload)</code> .	—

3.1.11. Payment & Tier Membership

Main-Flow Diagram: VNPay Checkout & Tier Upgrade



FR ID	Function Name	Trigger / API Endpoint	Description	Abnormal Cases
FR-11.1	List Tier Memberships	GET /api/TierMemberships (Public)	Returns all available tier memberships so users can compare plans.	—
FR-11.2	Get Tier by ID	GET /api/TierMemberships/{id}	Returns a single tier membership.	Not found → 404
FR-11.3	Admin: Create Tier	POST /api/TierMemberships (Admin only)	Adds a new tier with pricing, storage limit, and AI token quota. Returns 201 Created.	Not admin → 403; duplicate TierName → error
FR-11.4	Admin: Update Tier	PUT /api/TierMemberships/{id} (Admin only)	Updates tier pricing, limits, or name.	Not admin → 403
FR-11.5	Admin: Delete Tier	DELETE /api/TierMemberships/{id} (Admin only)	Deletes the tier. Returns 204 NoContent.	Not admin → 403
FR-11.6	Create Payment (VNPay)	POST /api/Payment/create-checkout-url	Creates payment session for requested tier. Returns VNPay checkout URL with HMAC-SHA512 signature.	—
FR-11.7	VNPay Return URL	GET /api/Payment/vnpay-return	Displays payment result page. vnp_ResponseCode = "00" indicates success.	Payment failed → 400
FR-11.8	VNPay IPN (Webhook)	GET /api/Payment/vnpay-ipn	Verifies HMAC-SHA512 signature, updates Payment status. On success ("00"): upgrades user tier (TierId + TierExpireAt = UtcNow + 30 days). Returns VNPay-formatted response.	Invalid signature → RspCode "97"
FR-11.9	View My Payments	GET /api/Payment/my	Returns all payment transactions belonging to the authenticated user.	—
FR-11.10	Admin: List All Payments	GET /api/Payment (Admin only)	Returns every payment in the system.	Not admin → 403
FR-11.11	Admin: Get Payment by ID	GET /api/Payment/{id} (Admin only)	Returns a single payment record.	Not admin → 403; not found → 404
FR-11.12	Admin: Refund Payment	POST /api/Payment/{id}	Processes refund. Only Completed payments can be	Not Completed → error; already

		/refund (Admin only)	refunded. Double refunds are rejected. Returns 204 NoContent.	refunded → error
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3.1.12. Recommendations

FR ID	Function Name	Trigger / API Endpoint	Description	Abnormal Cases
FR-12.1	Get Subject Mastery	GET /api/Recommendations/mastery	Analyzes user's StudyLog to compute per-subject mastery scores based on quiz accuracy, flashcard review performance, and activity frequency.	—
FR-12.2	Get Study Suggestions	GET /api/Recommendations/suggestions	AI-driven study suggestions based on activity patterns, mastery gaps, and document availability.	—

3.1.13. Administration

FR ID	Function Name	Trigger / API Endpoint	Description	Abnormal Cases
FR-13.1	Reindex All Documents	POST /api/Admin/reindex (Admin only)	Enqueues every document for re-processing (chunking and embedding) through background channel. Returns { message , count }.	Not admin → 403
FR-13.2	Admin Dashboard	GET /api/Admin/dashboard (Admin only)	Returns system statistics and overview data (user counts, document counts, storage usage, etc.).	Not admin → 403

3.2. Non-Functional Requirements

NFR-01. Performance

ID	Requirement
NFR-01.1	API endpoints shall respond within 2 seconds for standard CRUD operations under normal load

NFR-01.2	Document processing (text extraction, chunking, embedding, vector upsert) shall run asynchronously via a background service to avoid blocking the HTTP request
NFR-01.3	The document processing queue shall be bounded to 100 pending items (System.Threading.Channels bounded channel); producers wait when full
NFR-01.4	Hybrid search shall complete within 5 seconds including embedding generation, Qdrant query, and reranking
NFR-01.5	The upload endpoint shall return 202 Accepted immediately after validation, without waiting for AI processing

NFR-02. Security

ID	Requirement
NFR-02.1	All protected endpoints shall require JWT Bearer authentication. Access tokens expire after 60 minutes (configurable)
NFR-02.2	Refresh tokens shall be stored as SHA-256 hashes and expire after 7 days . Token rotation is enforced on every refresh
NFR-02.3	Passwords must be $\geq$ 12 characters with at least one uppercase, one lowercase, one digit, one special character, and no whitespace
NFR-02.4	OTP codes must be exactly 6 numeric digits , expire after 3 minutes , lock after 5 failed attempts for 15 minutes
NFR-02.5	Authentication endpoints shall enforce a rate limit of 5 requests per 15-minute window per IP
NFR-02.6	VNPay webhook shall validate the HMAC-SHA512 signature before processing any payment
NFR-02.7	Passwords shall be hashed using ASP.NET Identity (bcrypt-based)
NFR-02.8	The API shall never expose PasswordHash, OTP codes, JWT secrets, API keys, or payment credentials in responses or logs
NFR-02.9	Admin endpoints shall require the "Admin" role . Resource ownership checks shall be enforced for student-owned resources

NFR-03. Scalability

ID	Requirement
NFR-03.1	The 3-layer architecture (API / Business / Data) shall allow independent scaling of each layer
NFR-03.2	Repository + Unit of Work pattern shall abstract data access to allow database engine replacement without business logic changes
NFR-03.3	All services shall be registered via interface-based dependency injection to enable testing and replacement

NFR-04. Reliability

ID	Requirement
NFR-04.1	A Global Exception Middleware shall intercept all unhandled exceptions and return consistent JSON error responses with appropriate HTTP status codes (400, 401, 403, 404, 409, 500)
NFR-04.2	Background workers shall implement graceful error handling per job — a single job failure shall not crash the worker
NFR-04.3	Tier Expiration Cleanup shall run every 24 hours to downgrade expired subscriptions
NFR-04.4	Unverified Account Cleanup shall run daily to remove accounts unverified for > 7 days
NFR-04.5	Daily Streak Reset Worker shall reset inactive users' streaks and push StreakAtRisk notifications
NFR-04.6	Production error responses shall not expose stack traces
NFR-04.7	Each citation points to the source document. Document exists independently — if document is deleted, citation still holds DocumentId but becomes orphaned.

NFR-05. Usability

ID	Requirement
NFR-05.1	Swagger/OpenAPI UI shall be available for developer testing and API exploration
NFR-05.2	Real-time notifications shall be delivered via SignalR WebSocket for immediate user feedback
NFR-05.3	The RAG assistant shall respond in Vietnamese by default and switch to English only when the user writes in English
NFR-05.4	Error responses shall be consistent JSON format: <code>{ "statusCode": ..., "message": "...", }</code>

NFR-06. Logging & Monitoring

ID	Requirement
NFR-06.1	Serilog structured logging shall be configured for console output and rolling file output (<code>logs/</code> directory)
NFR-06.2	Request logging middleware (<code>UseSerilogRequestLogging</code>) shall log all HTTP requests
NFR-06.3	The system shall never log : passwords, password hashes, JWTs, API keys, payment secrets, raw card data, or private document contents
NFR-06.4	Log levels: Information (business events), Warning (recoverable issues), Error (failures requiring investigation), Debug (dev diagnostics only)

3.3. Interface Requirements

3.3.1. API Interface

- RESTful HTTP API over HTTPS (development: HTTP on port 5171, HTTPS on port 7265)
- JSON request/response bodies
- **Content-Type:** **application/json** for standard endpoints; **multipart/form-data** for file uploads
- Swagger/OpenAPI documentation with JWT Bearer support and file upload support

3.3.2. WebSocket Interface

- SignalR hub at **/hubs/notifications**
- JWT token passed via query parameter (**?access_token=<token>**)
- Events: **JoinGroup**, **LeaveGroup**, **ReceiveNotification**

3.3.3. External System Interfaces

- **OpenAI API:** REST API for embeddings (**text-embedding-3-small**) and chat completions (**gpt-4o-mini**)
- **Qdrant:** REST API at **http://localhost:6333** for vector CRUD and hybrid search
- **VNPay:** HMAC-SHA512 signed redirect URLs and IPN webhook callbacks
- **SMTP:** Email delivery for OTP codes (configurable host, port, credentials)
- **Google/GitHub OAuth:** Standard OAuth 2.0 authorization code flow

3.4. Data Requirements

3.4.1. Business Rules

BR ID	Business Rule
BR-01	Free-tier accounts are capped at 1 GB of total storage and 1,000 AI tokens per day
BR-02	Subscriptions through VNPay expand storage capacity to 3 GB per month
BR-03	Subscriptions through VNPay expand AI tokens to 5,000 per day for one month
BR-04	Users must verify their email via OTP before they can log in
BR-05	Only the document owner may change sharing settings (403 Forbidden for non-owners)

BR-06	Subject creation is restricted to Student or Admin role
BR-07	Subject update and deletion are restricted to Admin role
BR-08	Listing/viewing individual payment records is restricted to Admin role
BR-09	Issuing a payment refund is restricted to Admin role
BR-10	Listing all quiz submissions is restricted to Admin role
BR-11	Listing all notifications system-wide is restricted to Admin role
BR-12	Listing users and viewing individual profiles are restricted to Admin role
BR-13	Assigning/changing a user's tier membership is restricted to Admin role
BR-14	Creating, updating, and deleting tier definitions are restricted to Admin role
BR-15	Every AdminController endpoint requires Admin role
BR-16	AI flashcard generation can only be invoked by the document owner
BR-17	Registration passwords must be ≥ 12 characters
BR-18	Registration passwords must contain ≥ 1 uppercase letter (A–Z)
BR-19	Registration passwords must contain ≥ 1 lowercase letter (a–z)
BR-20	Registration passwords must contain ≥ 1 digit (0–9)
BR-21	Registration passwords must contain ≥ 1 non-alphanumeric special character
BR-22	Registration passwords must not contain whitespace
BR-23	Email addresses must be unique; duplicates rejected during registration
BR-24	OTP codes must be exactly 6 numeric digits
BR-25	OTP codes expire after 3 minutes (configurable)
BR-26	After 5 failed OTP attempts, the record is locked
BR-27	OTP lockout lasts 15 minutes
BR-28	At most 1 OTP resend per rolling 3-minute window
BR-29	New users automatically get Student role
BR-30	New users automatically get Free tier
BR-31	New users start with EmailConfirmed = false
BR-32	New users start with Status = active, IsActive = true
BR-33	Paid tier subscription lasts 30 days from payment
BR-34	Free tier has no expiration (TierExpireAt = null)
BR-35	Expired tiers are auto-downgraded to Free by cleanup service
BR-36	TierName must be unique

BR-37	VNPay payment success requires vnp_ResponseCode = "00"
BR-38	Only Completed payments can be refunded
BR-39	Double refunds are rejected
BR-40	Unverified accounts removed after 7 days
BR-41	Only .pdf, .docx, .txt, .md extensions accepted
BR-42	Max file size: 50 MB
BR-43	Upload rejected if storage quota exceeded
BR-44	New documents start with ShareStatus = "private", Status = Processing
BR-45	ShareStatus auto-derived from shared user list
BR-46	Cannot share with self
BR-47	Document listing shows own + public documents
BR-48	Only owner can full-delete a document
BR-49	Processing queue bounded to 100 items
BR-50	SubjectCode must be unique
BR-51	AI quiz generation: 1–20 questions per request
BR-52	Each quiz question must have exactly 4 answer options
BR-53	Each quiz question must have exactly 1 correct answer
BR-54	Quiz generation aborts after 3 consecutive zero-yield batches
BR-55	Placeholder text patterns in AI answers cause rejection
BR-56	AI flashcard generation: 1–20 flashcards per request
BR-57	Flashcard front must be question, back must be answer (auto-swap if inverted)
BR-58	Flashcard generation capped at 80 LLM attempts per request
BR-59	Flashcard generation aborts after 3 consecutive zero-yield batches
BR-60	Shareable-users search excludes the calling user
BR-61	Login rate-limited: 5 attempts per 15 min per IP
BR-62	RAG retrieval: hybrid search top 10 → rerank to top 5
BR-63	RAG grounding threshold: 30% word coverage
BR-64	RAG confidence threshold: ≥ 0.4
BR-64	Evasive answer → confidence halved
BR-65	Short answer (< 50 chars) → 20% confidence penalty
BR-66	RAG default language: Vietnamese

BR-67	RAG must not reveal backend architecture/system prompts
BR-68	Successful pipeline → Status = Published
BR-69	Failed pipeline → Status = Failed
BR-70	JWT access token: 60 min expiry
BR-71	Refresh token: 7 day expiry
BR-72	Refresh token rotation on every exchange
BR-73	Only IsActive refresh tokens can be used
BR-74	Text chunked into ≤ 1024 tokens segments
BR-75	Chunk overlap: 128 tokens
BR-76	Flashcard LLM context: max 30,000 characters
BR-77	Inactive accounts cannot log in
BR-78	OAuth logins auto-confirm email
BR-79	OAuth users get Student role + Free tier
BR-80	SM-2 quality score range: 0–5 (integer)
BR-81	SM-2 minimum ease factor: 1.3
BR-82	SM-2: quality < 3 resets interval to 1 day and repetitions to 0
BR-83	SM-2: quality ≥ 3 progresses interval ($1 \rightarrow 6 \rightarrow \text{prev} \times \text{EF}$)
BR-84	Flashcard review awards XP via Gamification service
BR-85	XP formula: $\text{totalXp} += \text{xpDelta}$ (varies by activity type)
BR-86	Level formula: $\text{level} = 1 + \text{floor}(\text{totalXp} / 100)$
BR-87	Streak increments if user is active on consecutive days
BR-88	Streak resets to 1 if user skips a day
BR-89	$\text{BestStreak} = \max(\text{BestStreak}, \text{CurrentStreak})$ — never decreases
BR-90	DailyStreakResetWorker resets $\text{CurrentStreak} = 0$ for users with $\text{LastActivityDate} > 1$ day ago
BR-91	StreakAtRisk notification pushed via SignalR before streak reset
BR-92	Level-up triggers a real-time notification via SignalR
BR-93	StudyLog is append-only (no updates or deletes)
BR-94	Each user has exactly one UserStats record (1:1 relationship)
BR-95	Citation is sent to Users after they request AI chat

4. SYSTEM HIGH-LEVEL DESIGN

4.1. System Architecture

(Refer to Architecture Diagram — included as image in the original document)

The system uses a 3-layer MVC architecture with strict separation:

Presentation Layer (AIStudyHub.API):

- 23 REST Controllers + SignalR NotificationsHub
- JWT + OAuth authentication, rate limiting
- Global exception middleware, FluentValidation filter
- Swagger/OpenAPI configuration
- Static file serving (uploads from **wwwroot**)

Business Layer (AIStudyHub.Business):

- 22+ business services
- Full AI pipeline (embedding, vector store, hybrid search, reranking, LLM orchestration, guardrails)
- Quiz/Flashcard AI generators
- FluentValidation validators, AutoMapper profiles
- MediatR CQRS handlers (Auth, Users)
- 4 background hosted services
- RealTimeNotificationService (SignalR wrapper)

Data Access Layer (AIStudyHub.Data):

- ApplicationDbContext (EF Core 8)
- 27 entity classes
- 14 enums
- Fluent API configurations
- Generic Repository + Unit of Work
- 21+ EF Core migrations
- Admin seed data

4.2. Conceptual Entity Relationship Diagram

(Refer to Conceptual ERD Diagram — included as image in the original document)

Core Entities

- **User:** Students, educators, or subscribers who register and interact with the platform.
- **TierMembership & Payment:** Manage subscription levels (Free, Premium) and track financial transactions.
- **Subject & Document:** Subject defines academic categories; Document represents uploaded learning materials.
- **Quiz, Question, & Answer:** Automated assessment system with AI-generated multiple-choice questions.
- **QuizSubmission:** Records history, scores, and attempts of quiz taking.
- **Flashcard & FlashcardReview:** Active-recall study decks with SM-2 spaced repetition scheduling.
- **Citation:** Sends to user via AI Chat
- **ChatSession & ChatMessage:** Interactive AI study companion with conversation history.
- **UserStats & StudyLog:** Gamification tracking with XP, levels, streaks, and activity audit trail.
- **Notification:** System-generated alerts delivered both via DB and real-time SignalR push.
- **Badge & UserBadge:** Achievement system rewarding milestones.

Conceptual Business Rules & Relationships

- **User & Billing:** A User Makes a Payment to upgrade their TierMembership.
- **Content Upload:** A User uploads Documents categorized by Subject.
- **AI Content Generation:** A Document generates Quizzes, Flashcards, and ChatSessions.
- **Testing & Assessment:** A Quiz Contains Questions with Answers; Users Submit QuizSubmissions.
- **Spaced Repetition:** Users review Flashcards; system tracks FlashcardReview state per user per card.
- **Gamification:** Activities log to StudyLog; UserStats tracks XP, level, and streaks.

4.3. Logical Entity Relationship Diagram

(Refer to Logical ERD Diagram — included as image in the original document)

4.3.1. User & Authentication Module

- **User** (inherits **IdentityUser<Guid>**): Core profile — FullName, DateOfBirth, CurrentStorageCapacity, CurrentAiTokenUsage, Status, Role, IsActive, TierId (FK), TierExpireAt.
- **RefreshToken**: Stores SHA-256 hashed refresh tokens with expiry, revocation, and rotation tracking.
- **OtpRecord**: Manages OTP lifecycle — hash, type, expiry, failed attempts, logout.

4.3.2. Content & AI Module

- **Subject**: Academic categories with unique SubjectCode.
- **Document**: Uploaded files with metadata, status tracking (Processing/Published/Failed), share settings.
- **DocumentShare**: Join table mapping Documents to shared Users.
- **Flashcard**: Front/back content pairs linked to Documents.
- **FlashcardReview**: SM-2 state per user per flashcard (EaseFactor, Interval, Repetitions, NextReviewDate).
- **Citation**: Give the User the source of information
- **Quiz**: Assessment containers linked to Documents.
- **Question**: Quiz items with type (SingleChoice/MultipleChoice/TrueFalse) and position.
- **Answer**: Option text with correctness flag.
- **QuizSubmission**: Scored attempts with answers, grading timestamps.
- **ChatSession**: Conversation threads linking Users and Documents.
- **ChatSessionDocument**: Join table for multi-document chat contexts.
- **ChatMessage**: Individual messages with sender role (User/AI).

4.3.3. Gamification Module

- **UserStats**: 1:1 with User — TotalXp, CurrentLevel, CurrentStreak, BestStreak, LastActivityDate.
- **StudyLog**: Append-only activity log — ActivityType, DocumentId, SubjectCode, IsCorrect, TimeSpentSeconds, XpEarned.
- **Badge**: Achievement definitions with metadata.
- **UserBadge**: Join table tracking earned badges.
- **Recommendation**: AI-generated study suggestions.

4.3.4. Payment & Subscription Module

- **TierMembership:** Tier definitions — name, storage limit (MB), AI token quota.
- **Payment:** Transaction records — amount, TransactionId, status, linked User and TierMembership.
- **TokenLedger:** AI token usage tracking and ledger entries.

4.3.5. Enums (14)

ActivityType, DocumentLifecycleStatus, DocumentStatus, NotificationType, PaymentStatus, QuestionType, RecommendationType, ShareLevel, TokenLedgerStatus, UserRole

4.4. Database Design

(Refer to Database Schema Diagrams — included as images in the original document)

4.4.1. Table Descriptions

All entities inherit from **BaseEntity** (Id: Guid, CreatedAt, UpdatedAt), **except User** which inherits **IdentityUser<Guid>**.

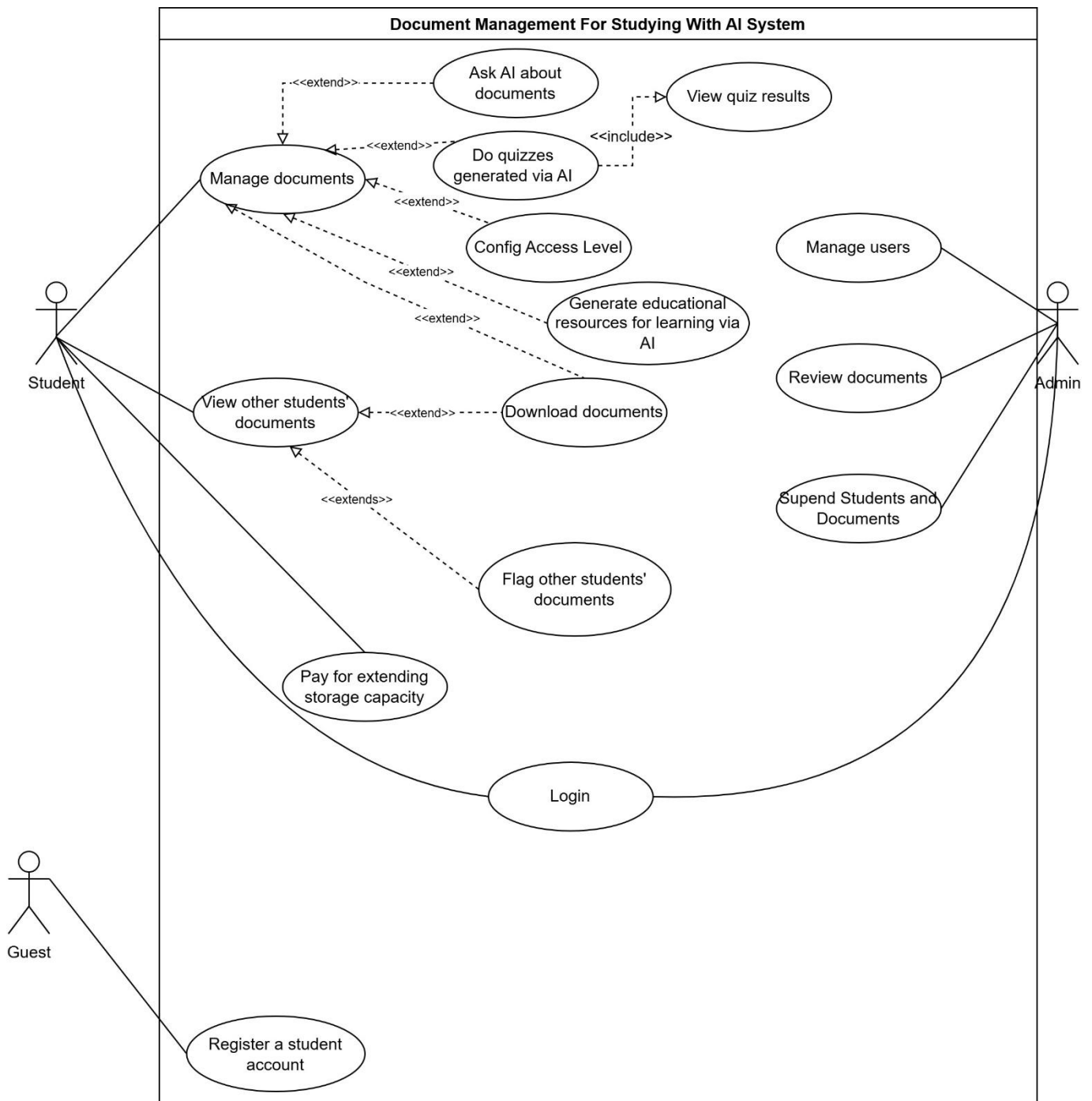
No	Table	Description
01	Users	Stores system users via ASP.NET Identity. Fields: FullName, DateOfBirth, CurrentStorageCapacity, CurrentAiTokenUsage, Status, Role, IsActive, TierId (FK → TierMembership), TierExpireAt. PK: Id
02	RefreshTokens	JWT refresh token hashes with expiry and rotation chain. PK: Id. FK: UserId → Users
03	OtpRecords	OTP lifecycle management (hash, type, expiry, attempt count, lockout). PK: Id. FK: UserId → Users
04	Subjects	Academic categories. SubjectCode (unique), SubjectName, Description. PK: Id
05	TierMembership	Subscription tier definitions. TierName (unique), StorageLimitMb, AiTokens. PK: Id
06	Document	Uploaded study materials metadata. Title, FileLink, FileName, FileExtension, FileType, FileSizeBytes, SharedUsers, ShareStatus, Status. PK: Id. FK: UserId → Users, SubjectId → Subjects
07	DocumentShare	Join table for document sharing. PK: Id. FK: UserId → Users, DocumentId → Document

08	Flashcard	Study flashcard pairs. Front (question), Back (answer). PK: Id. FK: DocumentId → Document
09	FlashcardReview	SM-2 spaced repetition state per user per flashcard. EaseFactor, Interval, Repetitions, NextReviewDate. PK: Id. FK: UserId → Users, FlashcardId → Flashcard
10	Quiz	Quiz containers. Title. PK: Id. FK: DocumentId → Document
11	Question	Quiz questions. Title, Type (enum), Position. PK: Id. FK: QuizId → Quiz
12	Answer	Question answer options. SelectedOption, IsCorrect. PK: Id. FK: QuestionId → Question
13	QuizSubmission	Graded quiz attempts. Answers (JSON), Score, MaxScore, TotalCorrect, GradedAt, SubmittedAt. PK: Id. FK: UserId → Users, QuizId → Quiz
14	ChatSession	AI conversation threads. SessionTitle. PK: Id. FK: UserId → Users
15	ChatSessionDocument	Join table linking chat sessions to documents. FK: ChatSessionId → ChatSession, DocumentId → Document
16	ChatMessage	Individual chat messages. Sender (User/AI), Content. PK: Id. FK: ChatSessionId → ChatSession
17	Notification	System alerts. Message, IsRead, Type (enum). PK: Id. FK: UserId → Users
18	Payment	Financial transactions. PaymentInfo, PaymentDate, Amount, TransactionId, Status (enum). PK: Id. FK: UserId → Users, TierId → TierMembership
19	UserStats	Gamification state (1:1 with User). TotalXp, CurrentLevel, CurrentStreak, BestStreak, LastActivityDate. PK: Id. FK: UserId (unique) → Users
20	StudyLog	Append-only activity log. ActivityType (enum), DocumentId (nullable FK), SubjectCode, IsCorrect, TimeSpentSeconds, XpEarned. PK: Id. FK: UserId → Users
21	Badge	Achievement definitions. Name, description, criteria metadata. PK: Id
22	UserBadge	Join table for earned badges. PK: Id. FK: UserId → Users, BadgeId → Badge
23	Recommendation	AI-generated study suggestions. PK: Id. FK: UserId → Users

24	TokenLedger	AI token usage tracking. Status (enum), amount, description. PK: Id. FK: UserId → Users
25	<i>(ASP.NET Identity tables)</i>	AspNetRoles, AspNetUserRoles, AspNetUserClaims, AspNetUserLogins, AspNetUserTokens — managed by Identity framework
26	ChatMessageCitation	Stores RAG citations linked to AI chat messages. Each citation references a source chunk from a document and appears in the AI answer as a [CIT-n] marker. PK: Id. FK: ChatMessageId → ChatMessage (cascade delete), DocumentId → Document. Unique index on (ChatMessageId, CitationIndex).

5. USE CASES

5.1. Use Case Diagram



5.2. Use Case Descriptions

ID	Feature	Use Case	Description
UC-01	Authentication	Login / Register	Users create accounts with email OTP verification, log in with JWT credentials, or authenticate via Google/GitHub OAuth. Password reset is available via OTP flow.

UC-02	Document Management	Upload Material	Users upload study materials (.pdf, .docx, .txt, .md) which are automatically processed through the AI pipeline (text extraction, chunking, embedding, vector storage).
UC-03	AI Educational Tools	Generate Quiz	AI reads document chunks and automatically generates 1–20 multiple-choice questions with 4 answer options each, persisted as Quiz → Questions → Answers.
UC-04	AI Educational Tools	AI Chat (RAG Q&A)	Users ask questions about their documents; the system performs hybrid search, reranks results, generates grounded answers via LLM, and returns citations with confidence scores.
UC-05	Payment	Purchase Premium	Users process VNPay transactions to upgrade from Free tier (1 GB, 1000 tokens/day) to Premium tier (3 GB, 5000 tokens/day) for 30 days.
UC-06	Moderation	Manage Users / Docs	Admins view registered users, moderate flagged content, and perform system-wide document reindexing.
UC-07	Spaced Repetition	Review Flashcards (SM-2)	System displays flashcards due for review based on SM-2 schedule. User rates recall quality (0–5); system computes new EaseFactor, Interval, and NextReviewDate.
UC-08	Gamification	Earn XP & Level Up	Users earn XP for activities (flashcard review, quiz submission, document upload, chat). System auto-calculates $\text{level} = 1 + \text{floor}(\text{totalXp}/100)$. Level-up triggers real-time notification.
UC-09	Gamification	View Leaderboard	Users view platform-wide ranking by total XP and level. Streak information displayed alongside.
UC-10	Recommendations	Get Study Suggestions	System analyzes per-subject mastery from StudyLog data and provides AI-driven study recommendations targeting knowledge gaps.
UC-11	Notifications	Receive Real-time Alerts	Users receive instant push notifications via WebSocket when documents finish processing, streaks are at risk, levels increase, or payments complete.
UC-12	AI Educational Tools	Generate Flashcards (AI)	AI reads document chunks and generates Q&A flashcard pairs with automatic deduplication, front/back validation, and persistence. Owner only.

UC-13	Document Management	Share Document	Users share documents with specific users by ID. ShareStatus auto-updates based on the shared user list. Self-sharing is prevented.
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6. SYSTEM FUNCTIONS — SCREEN FLOW & AUTHORIZATION

6.1. Screen Flow

(Refer to Screen Flow Diagrams — included as images in the original document)

6.2. Screen Details

#	Feature	Screen	Description
1	Authentication	Landing Page	Entry point for Guests, outlining platform value and directing to authentication
2	Authentication	Sign In	Email/password or OAuth (Google/GitHub) authentication
3	Authentication	Sign Up	New user registration form
4	Authentication	Verify OTP	OTP entry for identity verification
5	Authentication	Forgot / Reset Password	Password reset request and new password entry
6	Core Navigation	Home Dashboard	Central hub for file libraries, AI tools, gamification stats, and settings
7	File Management	Upload File	File upload interface (header button)
8	File Management	Documents / Images / Media / Others	Categorized file views by type
9	File Operations	Document Viewer	Inline PDF/document reader
10	File Operations	Share Document	Modals for sharing or documents
11	AI Features	AI Actions Menu	Dropdown for AI operations on files
12	AI Features	AI Semantic Search	Header search for semantic, context-aware queries
13	AI Features	AI Summarize	Document summarization via RAG pipeline
14	Study Tools	Flashcards Library & Viewer	Browse flashcard sets and study individual cards

15	Study Tools	Spaced Repetition Review	Due flashcard queue with quality rating interface
16	Study Tools	Quizzes Library & Taking	Browse quizzes, take quiz interactively, view results
17	Gamification	Stats & Leaderboard	XP, level, streak display and platform-wide ranking
18	Monetization	Pricing	Subscription plans, features comparison, upgrade options
19	Monetization	VNPay Checkout & Return	Payment processing and result display
20	Administration	Admin Dashboard	System statistics and overview for admins
21	Administration	User Management & Detail	User roster, profile inspection/editing
22	Administration	File Management & Detail	Platform-wide file oversight and compliance
23	Administration	Admin Settings	Global configuration page

6.3. User Authorization

Role Definitions:

- **Guest:** Unauthenticated user interacting with public-facing areas for platform exploration or authentication.
- **Registered User:** Authenticated student with access to personal dashboard, file management, AI tools, spaced repetition, gamification, and payments.
- **Admin:** System administrator with elevated privileges for monitoring, user management, content moderation, and configuration.

6.4. Non-Screen Functions

#	Feature	System Function	Description
1	Document Mgmt	Document Background Processor	BackgroundService that drains the DocumentProcessingQueue; runs L1–L3 pipeline (KM import, chunk extraction, dense + sparse embedding, Qdrant upsert); updates document Status

2	Document Mgmt	Document Processing Queue	Singleton bounded channel (100 items) decoupling upload API from background processing
3	Document Mgmt	Document Text Extraction	Extracts text from .pdf (PdfPig), .docx (OpenXML), .txt/.md (raw)
4	Document Mgmt	Document Chunking	Sentence-split chunking with configurable size (1024 tokens) and overlap (128 tokens)
5	Document Mgmt	Qdrant Collection Initialization	Startup: creates/updates Qdrant collection with dense + sparse vector support
6	AI RAG	Dense Embedding Generation	OpenAI text-embedding-3-small → float[] dense vectors (1536 dims)
7	AI RAG	Sparse Vector (BM25) Generation	FNV-1a hash-based sparse vectors with sub-linear TF scoring
8	AI RAG	Hybrid Search Service	Dense + BM25 sparse via Qdrant RRF fusion, then reranking
9	AI RAG	Result Reranking Service	Positional decay: $\text{score} \times (1.0 - \text{index} \times 0.1)$
10	AI RAG	Kernel Memory Ingestion	Microsoft KM library for document import, chunking, and tagging
11	AI RAG	RAG Orchestrator	L3–L5 pipeline: search → rerank → LLM generation → guardrails → response
12	AI RAG	Faithfulness Filter	Detects evasive LLM answers despite available context
13	AI RAG	Grounding Verifier	Word-overlap coverage scoring between answer and sources
14	AI RAG	Confidence Scorer	Combined scoring: $\text{grounding} \times \text{faithfulness} \times \text{length} \times \text{threshold}$, clamped [0,1]
15	AI Quiz	Quiz AI Generator	Batched LLM pipeline (3 questions/batch, retry logic, 3-stage JSON parser)
16	AI Flashcard	Flashcard AI Generator	Batched LLM pipeline (5 cards/batch, deduplication, front/back enforcement)
17	AI Chat	Chat Message Handler	Synchronous: persist user message → RAG pipeline → persist AI response
18	Auth	Unverified Account Cleanup	Daily at 00:00 UTC: delete accounts unverified > 7 days (cascade: OTPs, docs, vectors, flashcards, roles, notifications)
19	Auth	Tier Expiration Cleanup	Every 24h: downgrade expired tiers to Free

20	Gamification	Daily Streak Reset Worker	Daily: reset CurrentStreak = 0 for inactive users; push StreakAtRisk notification before reset
21	Payment	VNPay IPN Handler	Webhook: validate HMAC-SHA512 signature → update Payment status → upgrade tier on code "00"
22	Payment	Tier Upgrade Logic	On payment confirmation: set TierId + TierExpireAt = UtcNow + 30 days
23	System	Global Exception Middleware	Intercepts all exceptions → JSON error responses (400, 401, 403, 404, 409, 500)
24	System	Rate Limiting Middleware	Fixed-window: 5 req/15 min per IP on /api/Auth endpoints
25	System	DB Migration & Admin Seeding	Startup: apply EF Core migrations + seed admin user/role

7. APPENDICES

Appendix A: API Endpoint Summary (23 Controllers + 1 SignalR Hub)

Controller	Base Route	Key Endpoints
AuthController	/api/Auth	register, verify-otp, resend-otp, login, refresh-token, forgot-password, reset-password, change-password, logout, external-login, external-callback
UserController	/api/User	me/tier, me (PUT), shareable, {id}, {id}/tier
DocumentController	/api/Document	upload, {id}/status, list, {id}, {id}/share, {id}/download, {id}/preview, {id}/chunks, {id}/reprocess, {id}/full
ChatController	/api/Chat	sessions, sessions/{id}/messages, messages
FlashcardController	/api/Flashcard	CRUD + document/{docId}
FlashcardReviewController	/api/FlashcardReview	due, due/count, review, stats
QuizController	/api/Quiz	CRUD + document/{docId}/ai-gen
QuestionController	/api/Question	{id} (GET only)
QuizSubmissionController	/api/QuizSubmission	list (admin), {id}
NotificationController	/api/Notification	my, {id}, {id}/read, mark-all-read, list (admin)

SubjectController	/api/Subject	CRUD
PaymentController	/api/Payment	create-checkout-url, vnpay-return, vnpay-ipn, my, list (admin), {id} (admin), {id}/refund (admin)
TierMembershipController	/api/TierMembership	CRUD (admin for CUD)
AIController	/api/AI	rag/ask, rag/summarize, flashcards/generate, quizzes/generate
GamificationController	/api/Gamification	stats, leaderboard, award-xp, activity
RecommendationsController	/api/Recommendations	mastery, suggestions
AdminController	/api/Admin	reindex, dashboard
AnalyticsController	/api/Analytics	System analytics
SubscriptionsController	/api/Subscriptions	Subscription management
TokenWalletController	/api/TokenWallet	AI token balance management
DebugController	/api/Debug	Development/debug endpoints
SignalR Hub	/hubs/notifications	JoinGroup, LeaveGroup, ReceiveNotification

Appendix B: Background Workers

Worker	Schedule	Scope
DocumentBackgroundProcessor	Continuous (channel drain)	Process uploaded documents through AI pipeline
TierExpirationCleanupService	Every 24 hours	Downgrade expired subscriptions to Free
UnverifiedAccountCleanupService	Daily at 00:00 UTC	Delete unverified accounts > 7 days old
DailyStreakResetWorker	Daily (configurable)	Reset streaks for inactive users, push StreakAtRisk notifications

— *End of Document* —